

CompTIA Network+

N10-009

Study Notes

Based on Professor Messer video series

professormesser.com

What It Is

- Open Systems Interconnection Reference Model
- Broad overview of how data traverses networks — not a detailed specification
- NOT the OSI protocol suite — most protocols today are TCP/IP-based, but OSI model applies to them
- Many protocols can operate at each individual layer (tens or hundreds per layer)
- Common language across IT organizations

♦ SUPPLEMENTARY ♦

Data unit names by layer:

Layer	Unit
1 Physical	Bits
2 Data Link	Frames (packet + MAC header + trailer)
3 Network	Packets
4 Transport	Segments (TCP) / Datagrams (UDP)
5–7	Data

Fragmentation at L3 = splitting packets too large for the next network's MTU (Maximum Transmission Unit).

The Seven Layers

Layer	Name	Mnemonic (top-down)	Mnemonic (bottom-up)
7	Application	All	Abbreviations
6	Presentation	People	Pointless
5	Session	Seem	Students
4	Transport	To	Teach
3	Network	Need	Not
2	Data Link	Data	Do
1	Physical	Processing	Please

Layer Details

Layer 1 — Physical

- Physical signals: cables, fiber, wireless
- No real protocols — just getting a signal from A to B
- Problems: bad cable, bad fiber, wireless interference
- Troubleshooting: loopback tests, cable/fiber testing, checking adapter cards

Layer 2 — Data Link

- Fundamental communication between two devices
- MAC address layer (Media Access Control) — hardware address of adapter card
- Also called DLC (Data Link Control) layer
- Ethernet adapters, wireless adapters
- Switches forward traffic based on destination MAC address → "switching layer"

- Address types: MAC, EUI-48, EUI-64

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EUI = Extended Unique Identifier — a hardware address burned into your network card.

EUI-48 (48 bits = 6 bytes)

- This IS the standard MAC address
- Every Wi-Fi card, Ethernet port, etc. has one
- Format: 00:1A:2B:3C:4D:5E (six hex pairs)
- 48 bits = enough unique addresses for every device

EUI-64 (64 bits = 8 bytes)

- Longer version used by IPv6
- Device auto-generates it by taking its EUI-48 and inserting FF:FE in the middle
- No manual configuration needed

Why two? EUI-48 was designed for Ethernet/Wi-Fi. IPv6 needed a longer identifier, so EUI-64 was created. Both uniquely identify a network interface at Layer 2.

For the exam: MAC address = EUI-48 = EUI-64 = all Layer 2 hardware addresses.

Layer 3 — Network

- Routing layer — routers determine forwarding based on destination IP (Internet Protocol) address
- IP addresses, subnet masks, routing
- Fragmentation: splitting frames/packets into smaller pieces for networks requiring smaller frame sizes, reassembling on the other side

Layer 4 — Transport

- "Post office layer" — getting data from A to B
- Protocols: TCP (Transmission Control Protocol), UDP (User Datagram Protocol)
- Segmentation: breaking large data into smaller pieces, reassembling at destination
- Identified by TCP/UDP port numbers

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TCP vs UDP

	TCP	UDP
Connection	Yes (handshake first)	No
Reliability	Guaranteed delivery, ordered	Best-effort, no guarantees
Speed	Slower	Faster
Use cases	Web, email, file transfer	Video streaming, DNS, gaming

TCP = registered letter with tracking. UDP = shouting across a room.

Layer 5 — Session

- Communication management between endpoints
- Session initiation, stopping, restarting

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Endpoint: any device that sends or receives network traffic (e.g., laptop, server, phone).

- Control protocols, tunneling

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Control protocols manage ongoing communication state (e.g., RPC, NetBIOS). **Tunneling** encapsulates one protocol inside another to create a virtual point-to-point link (e.g., PPTP wraps traffic for VPN transport).

Layer 6 — Presentation

- Puts data into human-readable format
- Character encoding
- Application encryption/decryption (SSL/TLS)
- Often discussed together with layer 7

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SSL = Secure Sockets Layer. **TLS** = Transport Layer Security (modern successor). TLS is the current standard; “SSL” is still used colloquially. In the OSI model, SSL/TLS spans L5–L7 (session management, encryption, and application data encapsulation).

Layer 7 — Application

- What you see on screen — direct user interaction
- Examples: HTTP, HTTPS, FTP, DNS, POP3
- Thousands of application protocols

Real-World Mapping Summary

Layer	What You’ll See
7	Application interaction (Gmail, web browser, etc.)
6	SSL/TLS encryption/decryption
5	Control protocols, tunneling, session start/stop
4	TCP/UDP port numbers
3	IP addresses, subnet masks, routing
2	Ethernet frames, MAC addresses, EUI, switching
1	Cables, fiber optic, wireless signals

Wireshark Protocol Decode Example

► EXAMPLE

Captured frame 88 (2,005 bytes) broken down:

1. **Frame info** (L1) — physical bytes on wire
2. **Ethernet II** (L2) — source/destination MAC addresses
3. **Internet Protocol** (L3) — source/destination IP addresses (e.g., 72.14.247.19 → gmail.l.google.com)
4. **TCP** (L4) — source/destination ports (dst 443)
5. **SSL** (L5/6/7) — session management, encryption/decryption, application data (Google Mail)

Key insight: Wireshark’s decode maps to OSI layers bottom-up (physical → application).

Device	OSI Layer	Primary Function
Router	3	Routes between subnets
Switch	2	Forwards by MAC address
Firewall/NGFW	3, 4 & 7*	Filters/controls traffic
IDS/IPS	3, 4 & 7*	Detects/blocks attacks
Load Balancer	4 & 7*	Distributes server traffic
Proxy	7*	Intermediary for requests
NAS	—	File-level network storage
SAN	—	Block-level network storage
Access Point	2	Bridges wireless and wired
WLC	—	Centralized AP management

*= inferred, not stated in source

Router

- Routes data between IP subnets — same data center or across the world
- OSI layer 3 (network layer) device — uses IP addresses to determine next hop
- Layer 3 switch = switch with routing functionality built in (layer 2 switch + layer 3 router in one device)
- Connects diverse network types: LAN/WAN, copper/fiber
- May have many different interfaces for different connection types

Switch

- Forwards traffic based on MAC addresses — OSI layer 2 (data link) device
- Operates mostly in hardware via ASICs (Application-Specific Integrated Circuits)
- Enterprise features:
 - Power over Ethernet (PoE) — delivers power on same wires as ethernet
 - Layer 3 routing functionality (layer 3 switch)

Firewall

- Traditional: filters traffic based on TCP/UDP port numbers
- Next-Generation Firewall (NGFW): identifies and controls applications traversing the network
- Additional capabilities:
 - VPN — encrypted site-to-site tunnels (firewall-to-firewall)
 - Layer 3 routing — sits at network ingress/egress
 - NAT (Network Address Translation)
 - Dynamic routing protocols
- Manages communication between inside (LAN) and outside (internet)

IDS / IPS

- IDS (Intrusion Detection System) — detects and alerts on inbound attacks
- IPS (Intrusion Prevention System) — detects and blocks
- Looks for known attack types: OS exploits, application vulnerabilities, buffer overflows, XSS
- Much of this functionality now integrated into NGFWs
- IPS preferred in enterprise — IDS only alerts

Load Balancer

- Distributes traffic across multiple physical servers
- Transparent to end users — appears as a single server
- Detects server failures, removes failed servers from rotation, maintains uptime
- Optimization features:
 - TCP offload — speeds internal server communication
 - SSL offload — handles encryption for servers
 - Caching — answers requests immediately without hitting backend servers
 - QoS (Quality of Service) — prioritizes certain traffic
 - Application-centric load balancing — routes specific pages to specific servers

Proxy

- Sits between user and internet, makes requests on user's behalf
- Receives responses, verifies no malicious content, forwards to user
- Features:
 - Caching — returns cached responses, skips internet
 - Access control — authentication for internet access
 - URL filtering and content scanning
- Types:
 - Explicit proxy — requires OS/application configuration
 - Transparent proxy — works invisibly, no client configuration needed

Network-Attached Storage (NAS)

- Centralized file storage on the network
- File-level access — must transfer entire file to read/modify, write entire file back
- Commonly on isolated high-bandwidth network

Storage Area Network (SAN)

- Block-level access — read/write only the changed blocks (like a local drive)
- More efficient than NAS for large files with small modifications
- Commonly on isolated high-bandwidth network

Access Point (AP)

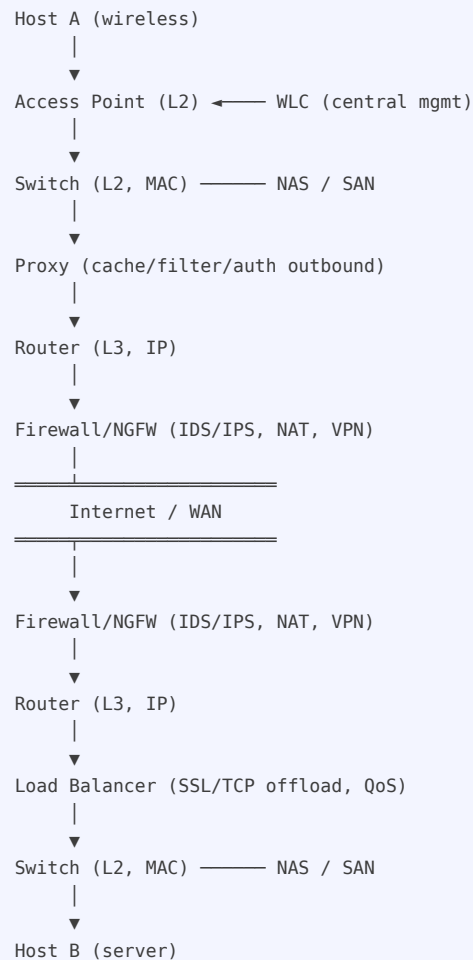
- Bridges wireless (802.11) and wired (802.3) networks
- OSI layer 2 device — translates between wireless and wired at data link layer
- NOT a wireless router — purpose-built for wireless communication only
- Enterprise environments use dedicated APs (not combo router/AP/switch like home)

Wireless LAN Controller

- Centralized management of all access points from one location (“single pane of glass”)
- Capabilities:
 - Deploy new APs with full configuration
 - Performance and security monitoring/alerting
 - Push configuration changes to all APs simultaneously
 - Usage reporting and capacity planning
- Enables seamless roaming between APs
- Typically proprietary — must match AP manufacturer

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Full data path (all devices inline):



Why this order:

1. **AP + WLC** — wireless client joins network
 2. **Switch** — L2 forwarding on local segment; NAS/SAN connect here
 3. **Proxy** — inspects/caches/filters outbound traffic
 4. **Router** — routes toward destination network
 5. **Firewall/NGFW (IDS/IPS)** — last checkpoint before WAN; NAT, VPN
 6. **WAN** — crosses internet
 7. **Firewall/NGFW (IDS/IPS)** — first checkpoint inbound
 8. **Router** — routes into internal network
 9. **Load Balancer** — distributes requests across servers
 10. **Switch** — delivers to server; NAS/SAN on same segment
- Proxy = client side (outbound). Load balancer = server side (inbound). NAS/SAN attach to switches, not inline.

CDN (Content Delivery Network)

- Distributes data efficiently from central point to end users
- Servers placed in geographical regions (North America, Asia, Europe, etc.)
- Caches content at each location — users hit nearest server
- Faster than routing all users to one central server
- Used by YouTube, streaming sites, small and large websites alike

VPN (Virtual Private Network)

- Connects to a remote network with all data encrypted
- Secure communication even across inherently insecure networks (public internet)
- **Concentrator / head-end device** — Central connection point for all VPN users
 - Purpose-built appliance for high-speed encryption/decryption in real time
 - Supports hundreds or thousands of simultaneous users
 - Often integrated into a next-generation firewall
- Smaller networks can use software-based VPN concentrators
- Client software: manufacturer-specific or built-in OS VPN clients (Windows, macOS, Linux)

QoS (Quality of Service)

- Prioritizes certain applications over others on the network
- Also called traffic shaping or packet shaping
- Controls bandwidth usage and data rates per application
- Example: real-time audio/video gets higher priority than file transfers
- Configured on firewalls, routers, or switches
- Devices may have pre-built application lists or allow custom entries

TTL (Time to Live)

- Timer that limits how long a packet persists on network
- Prevents infinite loops and stale data
- When TTL reaches zero, the task/packet is dropped

TTL in IP (hop count)

- Each router decreases TTL by 1 when processing a packet
- When TTL = 0, router discards the packet
- Defaults: macOS/Linux = 64 hops, Windows = 128 hops
- Typical internet path = 12–16 hops

- Located in the IPv4 header (visible in packet captures)
- Primary use: stops routing loops

Routing loops

- Router A thinks next hop is Router B, Router B thinks next hop is Router A
- Packet bounces back and forth indefinitely without TTL
- Traceroute shows loop: 10.1.10.1 → 10.2.10.2 → 10.1.10.1 → ...
- Often caused by a single static route misconfiguration

TTL in DNS (seconds)

- Defines how long a DNS resolution is cached
- Included in the DNS server response
- Example: `dig www.professormesser.com` → TTL 300 = cache for 5 minutes
- After TTL expires, client must re-query DNS
- Allows administrators to ensure IP changes propagate within TTL window

Cloud Deployment Types

- **Public cloud** — accessible to anyone on the internet
- **Private cloud** — internal use only, deployed in your own virtualized local data center
- **Hybrid cloud** — combination of public and private clouds for different applications

Software as a Service (SaaS)

- On-demand software accessed via browser login
- Someone else writes, manages, and upgrades the application
- Your data stored on a third-party cloud platform
- No local installation, no development work required
- Centralized management of application and data
- Complete application offering
- Examples: Google Mail, Office 365

Infrastructure as a Service (IaaS)

- Also called Hardware as a Service (HaaS)
- Cloud provides computing resources (hardware); you install and manage your own software
- You handle upgrades, data, access, and security
- Data in cloud but you have more control
- Example: web hosting where you buy server time

Platform as a Service (PaaS)

- Middle ground between SaaS and IaaS
- Cloud provider gives you tools/building blocks to develop your own application
- You manage development and customization
- Provider manages the underlying engine/platform that runs the application
- You maintain apps; provider maintains platform
- Example: Salesforce.com

Cloud Responsibility Matrix

Layer	SaaS	PaaS	IaaS	On-Prem
Information and Data	Customer	Customer	Customer	Customer
Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
Accounts and Identities	Customer	Customer	Customer	Customer
Identity and Directory Infra	Shared	Shared	Customer	Customer
Applications	Provider	Shared	Customer	Customer
Network	Provider	Shared	Customer	Customer
Controls				
Operating Systems	Provider	Provider	Customer	Customer
Physical Hosts	Provider	Provider	Provider	Customer
Physical Network	Provider	Provider	Provider	Customer
Physical Datacenter	Provider	Provider	Provider	Customer

Key takeaway: the more “as a service” you use, the less responsibility you take on — but also less control.

Cloud Computing Fundamentals

- Deploy applications and services with one click, seemingly unlimited resources
- **Elasticity** — scale up during high demand, scale down when demand drops
- Access applications from anywhere in the world
- **Multitenancy** — many customers share the same cloud infrastructure
- Efficiency in technology use and cost

Network Function Virtualization (NFV)

- When physical servers migrate to cloud, networking infrastructure virtualizes too
- Replaces physical routers/switches/firewalls with virtual
- Same functionality — managed from hypervisor

♦ SUPPLEMENTARY ♦

Hypervisor: software that creates/runs VMs on physical hardware, allocating CPU, memory, storage, and networking to each VM.

- Deploy new firewall/switch/router config instantly
- Flexibility for designing network connectivity and connecting cloud infrastructures globally

Virtual Private Cloud (VPC)

- Common application instance includes: web server, database server, load balancers, virtual switches/routers, virtual firewalls
- All run inside a VPC
- Larger environments use separate VPCs for different application instances or departments
- Keeps systems separated while managing them as individual virtual appliances

Transit Gateway

- Cloud router connecting multiple VPCs together
- Allows communication between different VPCs
- Central device that routes traffic between separate virtual private clouds

VPC Connectivity

VPN Connection

- Remote site or user workstation connects via VPN tunnel to transit gateway
- Provides access to private VPCs from outside

Internet Gateway (VPC Gateway)

- Makes application instances available to anyone on the internet
- Users anywhere in the world can access your applications

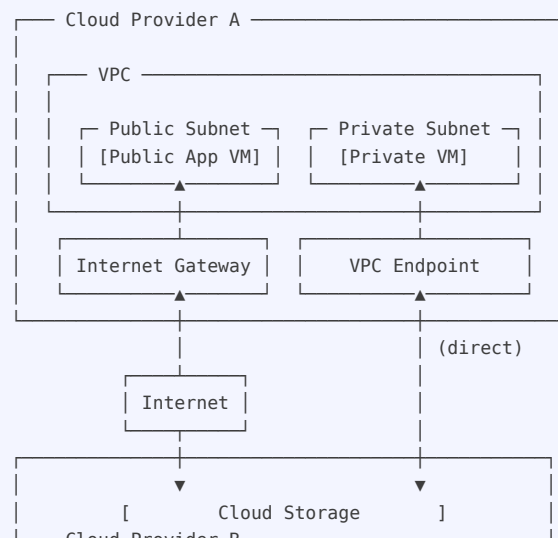
NAT Gateway (VPC NAT Gateway)

- NAT = Network Address Translation
- Allows VPC devices to reach the internet
- Translates private IP addresses to public IP addresses
- Security rules allow outbound to internet but can block inbound from external devices

VPC Endpoint

- Direct connection from a VPC on one cloud provider to a VPC on another cloud provider
- Used when organizations use multiple cloud providers but need connectivity between them
- Enables private subnet access without traversing the public internet

♦ SUPPLEMENTARY ♦



Public subnet: internet → internet gateway → public app.
Private subnet: VPC endpoint provides direct private connection, bypassing internet.

Network Security Lists

- Firewall rules applied at VPC level
- Control inbound and outbound traffic
- Based on port numbers and protocols (TCP/UDP)
- Can specify layer 3 addresses: individual IPs, CIDR blocks, IPv4/IPv6 ranges
- Once defined, applied to **all** virtual cloud networks
- Limitation: no granularity — same rules apply everywhere, even where not needed

Network Security Groups

- Assign rules to individual VNICs
- More granular than security lists — different rules per interface within the same subnet
- Trade-off: more granular but more admin overhead
- If even more security needed, consider a virtual firewall or other virtualized platform

FTP (File Transfer Protocol)

- Generic file transfer, cross-platform
- **TCP 20** — data transfer (active mode)
- **TCP 21** — control/commands
- Supports authentication (username/password)
- File operations: transfer, list, add, delete, rename

SSH (Secure Shell) — TCP 22

- Encrypted text-based remote terminal access
- All communication sent in encrypted form
- **SFTP (Secure FTP)** also uses TCP 22
 - File transfer encrypted via SSH protocol
 - Same file management capabilities as FTP
 - SSH is the underlying protocol for both remote terminal and secure file transfer

Telnet — TCP 23

- Telecommunication Network
- Non-encrypted remote terminal (looks like SSH)
- All data sent in the clear — including login credentials
- Rarely used today; replaced by SSH

SMTP (Simple Mail Transfer Protocol)

- **TCP 25** — server-to-server email transfer, plain text
- **TCP 587** — SMTP with TLS encryption
- Sending email and server-to-server transfer
- Receiving/managing inbox uses IMAP or POP3

DNS (Domain Name System)

- Translates FQDNs to IP addresses
- **UDP 53** — standard name queries
- **TCP 53** — large transfers (zone transfers, oversized responses)
- Critical — without it, no name resolution

DHCP (Dynamic Host Configuration Protocol)

- Automatic IP address configuration for network devices
- **UDP 67** — DHCP server
- **UDP 68** — DHCP client
- Pool of IP addresses assigned to connecting devices
- **Lease time** — IP address assigned temporarily; must renew at expiration

- **DHCP reservation** — permanently assigns IP to a specific MAC address
- Home: integrated into wireless router; enterprise: standalone servers

TFTP (Trivial File Transfer Protocol) — UDP 69

- Simple, fast file transfers with no authentication
- No directory listing or file management — just transfer
- Use case: VoIP phone boots via PoE → DHCP for IP → TFTP for config file
- Minimal overhead, very efficient

HTTP / HTTPS

- Hypertext Transfer Protocol
- **TCP 80** — HTTP, unencrypted web traffic
- **TCP 443** — HTTPS, encrypted via SSL/TLS
- Browser-based communication
- Most modern sites default to HTTPS (TCP 443)

NTP (Network Time Protocol) — UDP 123

- Synchronizes clocks across all network devices (OS, routers, switches)
- Critical for correlating log files across diverse devices
- 1ms accuracy between devices on same network
- Syncs multiple times per day automatically
- Configurable interval (hourly, daily, etc.)

SNMP (Simple Network Management Protocol)

- **UDP 161** — queries (management station → device)
- **UDP 162** — traps (device → management station, proactive alerts called SNMP traps)
- Used to monitor device performance, traffic, and status
- **Version 1** — single query/response, no encryption
- **Version 2** — bulk transfers, still no encryption
- **Version 3** — message integrity, authentication, and encryption

LDAP (Lightweight Directory Access Protocol)

- **TCP 389** — LDAP
- **TCP 636** — LDAP Secure (LDAPS)
- Database of network devices and users
- Hierarchical structure: Organization → Organizational Units (OUs) → Common Names (CNs)
- Easy to query and retrieve stored information

SMB (Server Message Block) — TCP 445

- Also called CIFS (Common Internet File System)
- Microsoft file/printer sharing and authentication
- Built into Windows (File Explorer), no extra software
- File shares, remote printing, file locking, permissions
- Older versions used NetBIOS; modern versions use TCP 445 directly

Syslog — UDP 514

- Standardized protocol for transferring logs to central location
- Used with SIEM
- Consolidates logs from routers/switches/firewalls/servers

MS-SQL (Microsoft SQL Server) — TCP 1433

- Microsoft's structured query language database server
- SQL provides standard language for querying/retrieving database data
- Other SQL databases use different port numbers

RDP (Remote Desktop Protocol) — TCP 3389

- View and control remote Windows desktops
- Clients available for Windows, macOS, Linux, iOS, Android

SIP (Session Initiation Protocol)

- **TCP 5060** — unencrypted
- **TCP 5061** — encrypted
- VoIP control protocol: initiates calls, disconnects sessions
- Also: video conferencing, IM, file transfers

Quick Reference

Protocol	Port(s)	Transport
FTP	20, 21	TCP
SSH/SFTP	22	TCP
Telnet	23	TCP
SMTP	25, 587	TCP
DNS	53	UDP/TCP
DHCP	67, 68	UDP
TFTP	69	UDP
HTTP	80	TCP
HTTPS	443	TCP
NTP	123	UDP
SNMP	161, 162	UDP
LDAP	389, 636	TCP
SMB	445	TCP
Syslog	514	UDP
MS-SQL	1433	TCP
RDP	3389	TCP
SIP	5060, 5061	TCP

Data Encapsulation

- Data is encapsulated within protocols, which are encapsulated within larger protocols
- Ethernet frame structure:
 - Ethernet Header → IP Header → TCP/UDP Header → Application Data → Ethernet Trailer
- Analogy: network = road, IP = truck, TCP/UDP = boxes, application data = items in boxes

♦ SUPPLEMENTARY ♦

Client
Server

Ethernet Header	Ethernet Payload			Ethernet Trailer
Ethernet Header	IP	IP Payload		Ethernet Trailer
Ethernet Header	IP	TCP	TCP Payload	Ethernet Trailer
Ethernet Header	IP	TCP	HTTP data	Ethernet Trailer

Each layer peels back to reveal the next level of encapsulation.

TCP (Transmission Control Protocol)

- **Connection-oriented** — formal setup and teardown of sessions
- **Reliable delivery** — receiving device acknowledges receipt of data
- Sender knows if data was properly delivered; can resend if no acknowledgment received
- Packets are numbered — receiver can request specific missing packets without full retransmission
- **Flow control** — receiver tells sender to speed up or slow down
- **Error recovery** — can retransmit lost portions

♦ SUPPLEMENTARY ♦

TCP establishes a “connection” (L4), not a “session” in the strict OSI sense. Sessions (L5) handle higher-level dialog management and checkpointing. The three-way handshake synchronizes sequence numbers at the transport layer.

UDP (User Datagram Protocol)

- **Connectionless** — no formal setup or teardown
- No acknowledgments sent back to sender

- **Unreliable** — not higher chance of failure, just no confirmation of delivery
- No error recovery or retransmission capability
- No flow control — receiver cannot tell sender to adjust speed
- Same probability of successful delivery as TCP; difference is lack of feedback

TCP vs UDP Summary

Feature	TCP	UDP
Connection	Oriented	Connectionless
Acknowledgments	Yes	No
Reliability	Reliable	Unreliable
Error recovery	Yes	No
Flow control	Yes	No
OSI layer	4 (Transport)	4 (Transport)

IP Addressing and Ports

- Destination IP address = which house (device) to deliver to
- TCP/UDP port number = which room (application) within the house
- **Socket** = IP address + protocol (TCP/UDP) + port number

♦ SUPPLEMENTARY ♦

A connection is uniquely identified by a **5-tuple**: source IP, source port, destination IP, destination port, and protocol. This is how a server handles thousands of clients on the same port — each connection differs by at least one element.

Port Numbers

- **Non-ephemeral ports** (0–1,023) — permanent, well-known server ports (e.g., 80, 443)
- **Ephemeral ports** (1,024–65,535) — temporary client ports, randomized per session
- Total range: 0–65,535
- TCP and UDP port numbers are independent — TCP 80 ≠ UDP 80
- Not a hard rule: servers can use ephemeral range, clients can use non-ephemeral

Multiplexing

- Multiple applications sent simultaneously between same devices using different port numbers
- OSI Layer 4 enables this via unique port numbers per application

Port Numbers and Security

- Changing port numbers is NOT a security mechanism
- Still need a firewall to allow/disallow traffic
- Changing well-known ports breaks client expectations (clients must be reconfigured)

Practical Example

► EXAMPLE

- Server 10.0.0.2 runs: web (TCP 80), VoIP (UDP 5004), email (TCP 143)
- Client 10.0.0.1 uses random ephemeral source ports for each session
- Response traffic reverses source/destination ports and IPs
- All three applications communicate simultaneously via different port numbers

Unicast

- One-to-one communication — one station sends directly to another
- No other device on the network is involved in the conversation
- Used for: web browsing, file transfers, email
- Works with both IPv4 and IPv6
- Disadvantage: to send same data to many devices, must create separate one-to-one sessions for each

Multicast

- One-to-many-of-many — sends to multiple subscribed recipients simultaneously
- Recipients subscribe to the multicast feed
- Used for: multimedia streaming, stock exchange data, routing updates
- Requires network equipment that understands multicast
- Not typically used across different networks or very large networks
- Works with both IPv4 and IPv6

Anycast

- One-to-one-of-many — single destination IP address routed to the closest of many devices
- Receiving devices are all configured similarly
- Traffic goes to whichever device is closest (by routing metric)
- Example: anycast DNS — query goes to nearest data center
- Works with both IPv4 and IPv6

Broadcast

- One-to-all — single packet sent to every device on the local network
- Scope limited to the local broadcast domain (cannot cross routers)
- Used for: routing updates, ARP requests
- IPv4 only — IPv6 replaced broadcast with multicast

ICMP (Internet Control Message Protocol)

- Protocol for checking if devices are alive and operating on the network
- Carried by IP but does NOT use TCP or UDP — it's its own protocol
- Most common use: ping command sends ICMP message, waits for response
- Also provides network feedback:
 - Destination network unreachable
 - Time exceeded (TTL expired)

GRE (Generic Routing Encapsulation)

- Creates tunnels between two endpoints (commonly used with VPNs)
- Encapsulates data within an IP packet, sends across tunnel, decapsulates on other side
- Does NOT provide encryption — need additional VPN protocols for that

VPN Concentrators

- Purpose-built appliance providing encryption/decryption at a central point
- Often standalone unit or integrated into existing firewall
- Usually hardware with specialized encryption chips for efficiency/throughput
- Can also be software-based for limited users
- Site-to-site VPN: connects corporate network to remote site over public internet using VPN concentrators (firewalls/routers) at each end

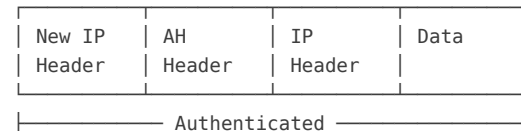
IPSec (Internet Protocol Security)

- One of the most popular protocols for encrypting VPN traffic
- Provides:
 - Encryption (confidentiality)
 - Digital signatures per packet (integrity)
 - Anti-replay functionality
- Standard protocol — works across different manufacturers' devices

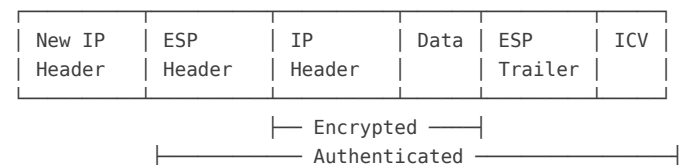
IPSec Protocols

Protocol	Name	Function
AH	Authentication Header	Integrity/authentication only, no encryption (data sent in clear with hashing)
ESP	Encapsulation Security Payload	Encrypts data AND provides authentication

AH — IP Packet with Authentication (tunnel mode):



ESP — IPSec Datagram with ESP (tunnel mode):



- ICV = Integrity Check Value (hash using MD5, SHA-1, or SHA-2)

IKE (Internet Key Exchange)

- Series of steps performed before sending data to establish the tunnel
- Both sides agree on encryption/decryption keys and parameters
- Agreement = Security Association (SA)

Phase 1

- Uses Diffie-Hellman to create shared secret key
- Operates over UDP port 500
- Called ISAKMP (Internet Security Association and Key Management Protocol)

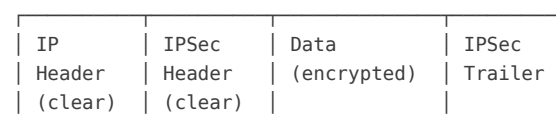
Phase 2

- Negotiates ciphers, key sizes
- Negotiates inbound and outbound Security Associations
- Encrypted data sent over ESP tunnel

Transport Mode vs Tunnel Mode

Transport Mode:

- Inserts IPSec header between original IP header and data
- Encrypts only the data portion
- Original IP header remains in the clear (destination visible to attackers)



Tunnel Mode:

- Original IP header AND data are all encrypted
- Adds new IP header with VPN concentrator as destination
- Attacker cannot see original destination

New IP Header (clear)	IPSec Header (clear)	Orig IP Hdr + Data (encrypted)	IPSec Trailer
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- Most implementations use tunnel mode for highest security

♦ SUPPLEMENTARY ♦

IPSec Protocol × Mode Matrix — any combination is valid:

	Transport Mode	Tunnel Mode
AH	Authenticates payload + most of original IP header	Authenticates entire original packet (new IP header wraps it)
ESP	Encrypts/authenticates payload only; original IP header exposed	Encrypts/authenticates entire original packet; new IP header wraps it

- **Mode** = what gets protected (transport = payload only, tunnel = entire original packet)
- **Protocol** = how it's protected (AH = integrity only, ESP = encryption + integrity)
- Most common in practice: **ESP + Tunnel mode** (site-to-site VPNs)
- AH rarely used alone since ESP provides everything AH does plus encryption
- Transport mode appears in host-to-host scenarios (e.g., L2TP/IPSec where L2TP provides the tunnel)

WebSocket

♦ SUPPLEMENTARY ♦

A persistent, full-duplex TCP connection between client and server. Unlike HTTP (request → response → close), a WebSocket stays open so either side can send data at any time without re-establishing the connection. Uses port 80 (ws:

Twisted Pair

- Most common cable type for wired Ethernet
- Multiple wires twisted together in a single sheath
- Wires paired together: transmit+/transmit-, receive+/receive-
- Same signal sent in opposite forms to detect and correct interference
- One wire always moving away from interference due to the twist
- Different pairs have different twist rates
- Cable itself has no speed — the Ethernet standard determines throughput
- IEEE 802.3 standards specify minimum cable category for each Ethernet standard

Cable Categories

- Cables classified by category (e.g., Cat 5, Cat 6, Cat 7)
- Ethernet standard specifies minimum category required
- Example: 1000BASE-T requires minimum Cat 5
- Any category at or above minimum will work

Coaxial Cable

- Two or more forms share a common axis
- Structure (inside out): conductor, insulator, shielding, jacket
- The outer shielding is the second conductor (return path), not just protection
- RG-6: common coax type for cable modem / internet connections

Twinaxial (Twinax)

- Two conductors inside the cable
- Associated with 10 Gbps Ethernet
- Often installed as part of SFP+ interface
- Full duplex over a single cable
- Limited to 5 meters
- Lower cost than fiber optics
- Lower latency than twisted pair

Plenum Cabling

- Plenum = space between drop ceiling and actual ceiling
- Shared air space where network cables, power, sensors may run
- If no ductwork, air flows through the plenum → concern in fires

Cable Jacket Types

Type	Material	Use
Standard	PVC (Polyvinyl Chloride)	Non-plenum
Plenum-rated	FEP (Fluorinated Ethylene Polymer) or low-smoke PVC	Plenum areas

- Plenum-rated cable produces less smoke and fewer hazardous fumes in a fire
- Plenum-rated cable is less flexible than standard cable
- Must use plenum-rated cable when installing in shared air spaces

RJ11 (Registered Jack type 11)

- 6-position, 2-conductor (6P2C)
- Used for analog telephone and DSL connections
- Smaller than RJ45

RJ45 (Registered Jack type 45)

- 8-position, 8-conductor (8P8C)
- Used for Ethernet connections
- Slightly larger and wider than RJ11

F-Connector

- Used for coaxial cable modem connections
- Threaded connector — screws in securely
- Single copper conductor on inside of cable carries the signal
- Also referred to as DOCSIS connector (Data Over Cable Service Interface Specification)
- Connects cable television infrastructure to cable modem

BNC (Bayonet Neill-Concelman)

- Bayonet connector — push in and twist to lock
- Named after Paul Neill (Bell Labs) and Carl Concelman (Amphenol)
- Used for WAN connections and other coax
- Twist to lock prevents accidental removal — must untwist to remove

Summary

Connector	Positions/ Conductors	Use	Lock Mechanism
RJ11	6P2C	Analog telephone, DSL	Clip
RJ45	8P8C	Ethernet	Clip
F-type	Single conductor	Cable modem (coax)	Threaded screw
BNC	Single conductor	WAN / coax	Bayonet twist

Overview

- Most popular networking standard in the world
- Varies in speed, cabling type, connectors, and equipment
- Supports both twisted-pair copper and fiber-optic media
- Standards managed by IEEE 802.3 committee

Standard Naming Convention

► EXAMPLE

1000BASE-T

Component	Meaning
1000	Speed in Mbps (1000 Mbps = 1 Gbps)
BASE	Baseband signaling (single frequency)
T	Media type (T = twisted-pair, F/SX = fiber)

- Number prefix = speed (e.g., 10G = 10 Gbps)
- BASE = baseband (single frequency), contrast with broadband (multiple frequencies)
- Suffix letters hint at media type but are not guaranteed to describe it

Common Standards

Standard	Speed	Media
1000BASE-T	1 Gbps	Twisted-pair copper
10GBASE-T	10 Gbps	Twisted-pair copper
1000BASE-SX	1 Gbps	Fiber optic (short wavelength)

Key Points

- IEEE naming conventions are informational but not strict — must read the full standard for exact details
- Baseband = single frequency for data transmission
- Broadband = multiple frequencies for data transmission
- SX suffix = short wavelength fiber optic

SC (Subscriber Connector)

- Also called: square connector, standard connector
- Push-in with snap lock — won't accidentally slip out
- Pull to unlock and remove
- Often combined in pairs (transmit + receive)

LC (Local Connector)

- Also called: Lucent connector, little connector
- Slightly smaller than SC
- Clip on top locks it in place — push down clip to release
- Often combined in pairs (transmit + receive)

ST (Straight Tip)

- Bayonet connector — push in and twist to lock
- Reverse twist to unlock and remove
- Protective ferrule around the fiber
- Prevents accidental dislodging in crowded racks

MPO (Multi-fiber Push On)

- 12 individual fibers in a single connector
- More efficient use of space than individual fiber connectors
- Push-in lock similar to SC — pull slightly to unlock
- Also called MTP (name by Corning) / MTP MPO

Summary

Connector	Full Name	Lock Mechanism	Key Feature
SC	Subscriber Connector	Push-snap	Square shape, common in DCs
LC	Local Connector	Top clip	Smaller than SC
ST	Straight Tip	Bayonet (push-twist)	Ferrule, twist lock
MPO	Multi-fiber Push On	Push-snap (like SC)	12 fibers in one connector

Overview

♦ SUPPLEMENTARY ♦

A transceiver is a small module (not a cable) that plugs into a switch/router port. The cable then plugs into the transceiver — it's the middleman between the switch and the cable.

- Transceiver = transmitter + receiver combined in one component
- Provides modularity — slide in the appropriate transceiver for your media/network type
- Each interface on a switch can have a different media type
- Transceiver must match the network type (Ethernet transceiver for Ethernet, Fibre Channel for Fibre Channel)
- Modularity comes at additional cost

Transceiver Types

Type	Full Name	Speed	Channels
SFP	Small Form-factor Pluggable	1 Gbps	1
SFP+	Enhanced Small Form-factor Pluggable	Up to 16 Gbps	1
QSFP	Quad Small Form-factor Pluggable	4 Gbps	4 × SFP
QSFP+	Quad Enhanced Small Form-factor Pluggable	40 Gbps	4 × SFP+

Form Factors

- SFP and SFP+ share the same physical form factor
- QSFP and QSFP+ share the same physical form factor
- QSFP/QSFP+ is slightly larger than SFP/SFP+ but not four times the size
- Space efficiency: four links in less space than four individual SFPs

Key Points

- Available in both copper and fiber versions
- Can swap copper transceiver for fiber (and vice versa) without changing the switch
- 10 Gbps connections commonly use SFP+
- QSFP+ extends four separate links over a single fiber — cost benefit for equipment and media
- Data center racks are 19 inches wide with limited space → density matters

Overview

- Transmits data using light (LED or laser)
- Difficult to tap without detection → used on secure networks
- Not susceptible to radio frequency interference
- Longer distances than copper (kilometers vs 100 meters for copper)
- Many different standards and connectors — check device documentation

Cable Structure

Component	Description
Core	Highly reflective center — carries light
Cladding	Low-reflective layer surrounding the core
Buffer coating	Protective outer layer
Ferrule	Protective cover at connector end

- Light source (LED or laser) on one end, receiver on the other
- Light bounces through the reflective core

Multimode Fiber

- Short-range communication (up to 2 km)
- Uses inexpensive light source (LED)
- Larger core (larger than wavelength of light)
- Multiple modes (paths) of light propagate through the fiber

Single-Mode Fiber

- Long-range communication (up to 100 km)
- Uses intense LED or laser
- Smaller core — only one mode of light propagates
- No need to regenerate signal over long distances

802.11 Standards

- Managed by IEEE (Institute of Electrical and Electronics Engineers)
- All wireless LAN standards fall under 802.11
- Wi-Fi Alliance tests/certifies devices for interoperability (Wi-Fi trademark logo)

Generational Naming

Standard	Generation	Frequencies	Max Theoretical Link Rate
802.11a	(Wi-Fi 1)	5 GHz	6–54 Mbps
802.11b	(Wi-Fi 2)	2.4 GHz	1–11 Mbps
802.11g	(Wi-Fi 3)	2.4 GHz	6–54 Mbps
802.11n	Wi-Fi 4	2.4 GHz / 5 GHz	72–600 Mbps
802.11ac	Wi-Fi 5	5 GHz	433 Mbps – 6.9 Gbps
802.11ax	Wi-Fi 6 and 6E	2.4 GHz / 5 GHz / 6 GHz	574 Mbps – 9.6 Gbps
802.11be	Wi-Fi 7	2.4 GHz / 5 GHz / 6 GHz	1.4–46.1 Gbps

- 802.11a/b/g rarely used today; generational names are informal for those
- Newer standards use more antennas/radios → higher link rates
- Link rates are theoretical maximums

Mobile Networking

4G / LTE

- LTE = Long-Term Evolution
- Unified GSM and CDMA into a single standard
- Download speeds: 150 Mbps
- LTE-A (LTE Advanced): 300 Mbps

5G

- Introduced 2020
- Higher Frequencies
- Goal: up to 10 Gbps throughput
- Typical speeds: 100–900 Mbps
- Enables expanded IoT footprint (more data, faster notifications and monitoring, cloud processing)
- Comparable bandwidth to wired home internet

Satellite Networking

- Used for remote locations without terrestrial internet access
- Typical speeds: 100 Mbps down / 5 Mbps up

Latency

Technology	Latency
Traditional satellite	500 ms (250 up + 250 down)
Starlink	40 ms (goal: 20 ms)

Limitations

- Requires direct line of sight to satellite
- Rain fade — loss of connectivity during storms
- Higher cost and complexity than terrestrial options
- May need backup connectivity for weather outages

Three-Tier Architecture

- Very common enterprise network design
- Three layers: core, distribution, access

Core Layer

- Central point for all resources
- Contains servers, applications, databases, critical services
- Core routers connect resources together
- Analogous to “downtown” of a city

Distribution Layer

- Midpoint between users and core resources
- Series of switches providing redundancy and connectivity
- Connects access layer to the core
- Analogous to highways connecting neighborhoods to downtown

Access Layer

- Where users physically connect to the network
- Switches located close to users (same floor of a building)
- Connects to one or more distribution switches
- Analogous to local access roads

Redundancy

- Multiple links between access and distribution layers
- Often multiple links between distribution and core
- If one component fails, alternate paths complete communication

Multi-Building Campus

- Each building has access switches per floor
- Each building's access switches connect to distribution switches
- All distribution switches connect to the core (often in a central data center)

Collapsed Core Architecture

- Two-tier design — combines core and distribution into one layer
- Access layer remains the same
- More simplified design, easier troubleshooting
- Less expensive — fewer devices to purchase and install
- Less redundancy — not as resilient if a component fails

Data Center Traffic Flow

East-West Traffic

- Traffic origination and destination both within the same data center
- Example: file server sending data to an image server on the same network
- Better response times — all on the same local network

North-South Traffic

- Traffic entering or leaving the data center from/to an external source
- Example: traffic to/from the internet
- Different security posture — unknown external origins/destinations

Summary

Architecture	Layers	Use Case
Three-tier	Core, Distribution, Access	Large enterprise / campus
Collapsed core	Core+Dist combined, Access	Smaller organizations

Traffic Flow	Direction	Security Concern
East-West	Within data center	Lower — internal traffic
North-South	Into/out of data center	Higher — external source

Star (Hub-and-Spoke)

- Most popular topology in large networks
- Central device connects all other devices
- All communication passes through the central hub
- Example: switched Ethernet — switch in the middle, all devices connect to it

Mesh

- Each device/location connects via multiple network paths
- If one link fails, traffic uses an alternate path
- Enables load balancing — split traffic across multiple connections
- Most commonly deployed in WANs to maintain connectivity regardless of link availability
- Can be used in LANs but less common

Point-to-Point

- Single point connected to a single point
- Common in older WANs: point-to-point T1 or T3 connections
- Also used in LANs: connecting buildings on a campus

Hybrid

- Combines multiple topology types in one enterprise network
- Example: star in one part, point-to-point in another, mesh in a third
- Reflects reality of large networks — different sections have different needs

Spine and Leaf

- Used in data centers
- Spine switches at the top, leaf switches in the middle, devices at the bottom
- Every leaf connects to every spine
- Leaf switches do NOT connect directly to each other
- Spine switches do NOT connect directly to each other
- Top-of-rack switching: leaf switch at the top of each physical rack
 - All devices in rack connect to their rack's leaf switch
 - Cabling self-contained within each rack
 - Redundancy via multiple spine connections
 - Performance: never more than one switch hop from any other device

- Cost scales with number of racks — separate switch per rack

Summary

Topology	Description	Common Use
Star	Central device, all nodes connect to it	LAN (Ethernet)
Mesh	Multiple paths between nodes	WAN
Point-to-Point	Direct link between two endpoints	WAN, campus links
Hybrid	Mix of multiple topologies	Enterprise
Spine & Leaf	Two-tier non-interconnected switch layers	Data center

Fundamentals

- Binary uses two numbers: 0 and 1
- Each 0 or 1 is a **bit**
- 8 bits = 1 **byte** (also called an **octet**)

Conversion Chart

- Start from right with 1, double each position moving left
- 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1
- Chart can extend left: 256, 512, 1024, etc.
- Each position is a power of 2 ($2^0 = 1$, $2^1 = 2$, $2^2 = 4$, ...)

Binary to Decimal

1. Write binary value under the conversion chart
2. Where binary digit is 0 → value is 0
3. Where binary digit is 1 → take the chart number above it
4. Add all values together

► EXAMPLE

Binary	Calculation	Decimal
00000010	2	2
10000010	128 + 2	130
11111111	128+64+32+16+8+4+2+1	255

Decimal to Binary

1. Write conversion chart above empty 8-bit placeholders
2. Starting from left (128): is the running total ≤ target?
 - Yes → place 1, add that value to running total
 - No → place 0, move to next column
3. Repeat for each column until all 8 bits are filled

► EXAMPLE

154 Decimal

128	64	32	16	8	4	2	1
1	0	0	1	1	0	1	0

- $128 \leq 154$? Yes → 1 (running: 128)
- $128+64=192 \leq 154$? No → 0
- $128+32=160 \leq 154$? No → 0
- $128+16=144 \leq 154$? Yes → 1 (running: 144)
- $144+8=152 \leq 154$? Yes → 1 (running: 152)
- $152+4=156 \leq 154$? No → 0
- $152+2=154 \leq 154$? Yes → 1 (running: 154)
- $154+1=155 \leq 154$? No → 0

Result: 10011010

Bit Count and Possible Values

- With 8 bits, any value 0–255 can be represented
- Total possible values = 2^n (where n = number of bits)

Bits	Possible Values (2^n)
2	4
3	8
4	16
5	32
6	64
7	128
8	256

Why Subnet?

- Impossible for one device to know every other device's location
- Subnetting creates smaller networks
- Routers forward traffic between subnets to reach any destination

Variable-Length Subnet Masks (VLSM)

- Subnet masks not limited to classful boundaries (/8, /16, /24)
- Any number of bits can be used for the mask
- Network admin controls exact network size
- Analogy: cutting a pizza into however many slices you need

Subnet Mask Anatomy

[Network bits (default class)]	[Subnet bits (borrowed)]	[Host bits (remaining)]

- Network bits: default from the class (A=8, B=16, C=24)
- Subnet bits: borrowed from host portion beyond the default
- Host bits: whatever remains after network + subnet bits

◆ SUPPLEMENTARY ◆

Subnets are a design tool, not a protocol feature. Routers only see the total prefix length — they don't distinguish between "class network bits" and "borrowed subnet bits." A routing table entry is just prefix + length (e.g., 10.1.1.0/24). The class-based breakdown only matters to the admin deciding how to carve up address space. Once advertised, it's all just prefix length pointing to a next hop.

Formulas

- **Number of subnets** = $2^{\text{(subnet bits)}}$
- **Hosts per subnet** = $2^{\text{(host bits)}} - 2$
 - Subtract 2 for network address and broadcast address

Calculation Examples

► EXAMPLE

10.1.1.0/24 (Class A, default /8)

- Subnet bits borrowed: $24 - 8 = 16$
- Host bits remaining: $32 - 24 = 8$
- Subnets: $2^{16} = 65,536$
- Hosts per subnet: $2^8 - 2 = 254$

► EXAMPLE

192.168.11.0/26 (Class C, default /24)

- Subnet bits borrowed: $26 - 24 = 2$
- Host bits remaining: $32 - 26 = 6$
- Subnets: $2^2 = 4$
- Hosts per subnet: $2^6 - 2 = 62$

► EXAMPLE

172.16.55.0/21 (Class B, default /16)

- Subnet bits borrowed: $21 - 16 = 5$
- Host bits remaining: $32 - 21 = 11$
- Subnets: $2^5 = 32$
- Hosts per subnet: $2^{11} - 2 = 2,046$

Powers of 2 Reference

2^n	Value
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1,024
2^{11}	2,048
2^{12}	4,096
2^{16}	65,536

Overview

- Class-based subnetting not used since 1993
- Still referenced colloquially (class A, B, C)
- Classes serve as the starting point for subnetting calculations

IP Address Classes

Class	First Octet	First Bits	Default Subnet Mask	Network Bits	Host Bits
A	0–127	0	255.0.0.0	8	24
B	128–191	10	255.255.0.0	16	16
C	192–223	110	255.255.255.0	24	8
D	224–239	1110	(multicast)	—	—
E	240–255	1111	(reserved)	—	—

- Class D: multicast, not assigned to individual devices
- Class E: reserved, not used for any purpose

Network/Host Bit Split

Class A	255 11111111 [Net (8)]	0 00000000 [	0 00000000 Hosts (24)	0 00000000]
Class B	255 11111111 [	255 11111111 Network (16)	0 00000000 [	0 00000000 Hosts (16)
Class C	255 11111111 [	255 11111111 Network (24)	255 11111111 [	0 00000000 Host(8)]

Determining Class from IP Address

- Look at the first octet only
- Examples:
 - 17.22.90.7 → first octet 17 → Class A
 - 220.10.77.40 → first octet 220 → Class C
 - 165.245.0.1 → first octet 165 → Class B
 - 192.1.12.5 → first octet 192 → Class C

Four Important Subnet Values

Value	How to Calculate
Network address	Set all host bits to 0
First usable host	Network address + 1
Broadcast address	Set all host bits to 1
Last usable host	Broadcast address – 1

Calculation Examples

► EXAMPLE

Class A: 10.74.222.11 /8

- Network/host split: 10 | 74.222.11
- Network address: 10.0.0.0
- First host: 10.0.0.1
- Broadcast: 10.255.255.255
- Last host: 10.255.255.254

► EXAMPLE

Class B: 172.16.88.200 /16

- Network/host split: 172.16 | 88.200
- Network address: 172.16.0.0
- First host: 172.16.0.1
- Broadcast: 172.16.255.255
- Last host: 172.16.255.254

► EXAMPLE

Class C: 192.168.4.77 /24

- Network/host split: 192.168.4 | 77
- Network address: 192.168.4.0
- First host: 192.168.4.1
- Broadcast: 192.168.4.255
- Last host: 192.168.4.254

IP Address Fundamentals

- Unique identifier every device needs to communicate via IP
- Four decimal numbers separated by periods (e.g., 192.168.1.165)
- Each number = 1 octet = 8 bits
- Total: 32 bits (4 bytes / 4 octets)
- Maximum value per octet: 255
- Operates at OSI layer 3 (network layer)

Subnet Mask

- Also a 4-octet value (e.g., 255.255.255.0)
- Used with IP address to determine what subnet a device is on
- Not transmitted across the network — used locally to decide what's local vs. remote
- Determines which addresses are on the local subnet vs. outside it

Default Gateway

- IP address of the router on your local subnet
- Used to communicate outside your local subnet
- Must be an address on your local network
- Configured by the network administrator

Special Addresses

Loopback

- Refers to the local device without needing to know its IP
- Range: 127.0.0.1 through 127.255.255.254
- Used to confirm local IP stack is working properly

Reserved (Class E)

- Range: 240.0.0.1 through 254.255.255.254
- Set aside for future use or testing
- Should never be assigned to devices

Virtual IP Address (VIP)

- Not associated with a physical Ethernet adapter
- Assigned to internals of a device (VMs, router interfaces)
- Allows consistent reference to a device regardless of physical adapter

IP Address Configuration

Manual (Static)

- Historically required visiting each device to configure IP, subnet mask, default gateway
- Any network changes required revisiting the device

DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns IP address, subnet mask, default gateway
- Works on wired and wireless networks
- Enables plug-and-play network connectivity

APIPA (Automatic Private IP Addressing)

- Assigned when no DHCP server is available
- Link-local address — can only communicate on local subnet (no internet)
- Range: 169.254.0.1 through 169.254.255.254
- Functional range (assigned to devices): 169.254.1.0 through 169.254.254.255
- First and last 256 addresses are reserved
- Uses ARP to confirm no other device has the chosen address

Public vs. Private IP Addresses

- IPv4 has exhausted all available public address blocks
- Private addresses cannot be routed on the public internet
- NAT converts private → public for internet communication

RFC 1918 Private Ranges

Range	Addresses	Class	CIDR	Subnet Mask	Host Bits
10.0.0.0 – 10.255.255.255	16M+	1 × A	10.0.0.0/8	255.0.0.0	24
172.16.0.0 – 172.31.255.255	1M+	16 × B	172.16.0.0/12	255.240.0.0	20
192.168.0.0 – 192.168.255.255	165K+	256 × C	192.168.0.0/16	255.255.0.0	16

Classless Subnetting (CIDR)

- Replaced class-based subnetting in 1993
- Classless Inter-Domain Routing (CIDR)
- Subnet mask can be any length, not limited to /8, /16, /24
- CIDR block notation: /n where n = number of 1-bits in the mask

CIDR Block Notation

- /24 = 255.255.255.0
- /16 = 255.255.0.0
- /8 = 255.0.0.0
- Common format: 192.168.1.44/24 (IP address with mask length)

Where Each Notation Is Used

- Operating systems (e.g., Windows): expect decimal subnet mask
- Routers/switches: often expect CIDR block notation
- Check device documentation for preferred format

Subnet Mask Structure

- Contiguous 1's on the left, 0's on the right
- 1-bits = network portion
- 0-bits = host portion
- Never mixed (no 0 between 1's)

Binary-to-Decimal Subnet Octet Values

Binary	Decimal	CIDR bits in octet
00000000	0	0
10000000	128	1
11000000	192	2
11100000	224	3
11110000	240	4
11111000	248	5
11111100	252	6
11111110	254	7
11111111	255	8

Conversion Examples

► EXAMPLE

/12 = 255.240.0.0

- Binary: 11111111.11110000.00000000.00000000
- Network bits: 12 | Host bits: 20

► EXAMPLE

/16 = 255.255.0.0

- Binary: 11111111.11111111.00000000.00000000
- Network bits: 16 | Host bits: 16

► EXAMPLE

/19 = 255.255.224.0

- Binary: 11111111.11111111.11100000.00000000
- Network bits: 19 | Host bits: 13

► EXAMPLE

/20 = 255.255.240.0

- Binary: 11111111.11111111.11110000.00000000
- Network bits: 20 | Host bits: 12

► EXAMPLE

/26 = 255.255.255.192

- Binary: 11111111.11111111.11111111.11000000
- Network bits: 26 | Host bits: 6

The Subnetting Problem

Given 192.168.1.0/24, find optimal subnet mask for 40 devices per subnet:

Mask	CIDR	Subnets	Hosts/Subnet
255.255.255.0	/24	1	254
255.255.255.128	/25	2	126
255.255.255.192	/26	4	62
255.255.255.224	/27	8	30

Answer: /26 — smallest mask that still supports 40+ hosts per subnet.

Four Values to Calculate

1. **Network address (subnet ID)** — first address in the subnet
2. **Broadcast address** — last address in the subnet
3. **First usable host** — subnet ID + 1
4. **Last usable host** — broadcast – 1

Magic Number Method Steps

No binary-to-decimal conversions needed — uses simple subtraction and multiples.

1. Convert subnet mask to decimal (if not already)
2. Identify the **interesting octet** — the octet that is neither 255 nor 0
3. Calculate magic number: **256 – interesting octet value**
4. Determine host ranges (multiples of the magic number)
5. Find which range the IP falls into
6. Network address = first value in that range
7. Broadcast address = subnet ID + magic number – 1

CIDR-to-Interesting-Octet Reference

CIDR for	/9	/10	/11	/12	/13	/14	/15	/16
interesting octet								
2								
CIDR for	/17	/18	/19	/20	/21	/22	/23	/24
interesting octet								
3								
CIDR for	/25	/26	/27	/28	/29	/30		
interesting octet								
4								
Magic number	128	64	32	16	8	4	2	1
Subnet mask	128	192	224	240	248	252	254	255

/31 and /32 are not included — /31 leaves no room for hosts, /32 leaves 0 host bits.

Subnet ID Rules

- If mask octet = 255 → copy IP address octet down
- If mask octet = 0 → write 0
- If mask octet = interesting → use magic number to find range start

Broadcast Address Rules

- If mask octet = 255 → copy subnet ID octet down
- If mask octet = 0 → write 255
- If mask octet = interesting → subnet ID + magic number – 1

► EXAMPLE

165.245.77.14 / 255.255.240.0

Octet	Mask	IP	Subnet ID	Broadcast
1	255	165	165	165
2	255	245	245	245
3	240	77	64	79
4	0	14	0	255

- Interesting octet: 3 (mask = 240)
- Magic number: 256 – 240 = 16
- Ranges: 0–15, 16–31, 32–47, 48–63, **64–79**, 80–95...
- IP octet 3 = 77 → falls in 64–79
- Subnet ID: 165.245.64.0
- Broadcast: 165.245.79.255
- First host: 165.245.64.1
- Last host: 165.245.79.254

► EXAMPLE

10.180.122.244 / 255.248.0.0

Octet	Mask	IP	Subnet ID	Broadcast
1	255	10	10	10
2	248	180	176	183
3	0	122	0	255
4	0	244	0	255

- Interesting octet: 2 (mask = 248)
- Magic number: 256 – 248 = 8
- Ranges: 0–7, 8–15, ... **176–183**, ...
- IP octet 2 = 180 → falls in 176–183
- Subnet ID: 10.176.0.0
- Broadcast: 10.183.255.255
- First host: 10.176.0.1
- Last host: 10.183.255.254

► EXAMPLE

172.16.242.133/27

Octet	Mask	IP	Subnet ID	Broadcast
1	255	172	172	172
2	255	16	16	16
3	255	242	242	242
4	224	133	128	159

- Interesting octet: 4 (mask = 224, /27)
- Magic number: $256 - 224 = 32$
- Ranges: 0–31, 32–63, 64–95, 96–127, **128–159**, ...
- IP octet 4 = 133 → falls in 128–159
- Subnet ID: 172.16.242.128
- Broadcast: 172.16.242.159
- First host: 172.16.242.129
- Last host: 172.16.242.158

Overview

- No math involved — all values predefined in a chart created at the start
- Only arithmetic: adding/subtracting 1 for first/last IP
- Same structure as magic number method but eliminates the 256 – mask calculation

The Chart

CIDR (octet 1)	CIDR (octet 2)	CIDR (octet 3)	CIDR (octet 4)	Network	Address	Subnet Mask
/1	/9	/17	/25	2	128	128
/2	/10	/18	/26	4	64	192
/3	/11	/19	/27	8	32	224
/4	/12	/20	/28	16	16	240
/5	/13	/21	/29	32	8	248
/6	/14	/22	/30	64	4	252
/7	/15	/23		128	2	254
/8	/16	/24		256	1	255

- Networks column: start at 2, double down
- Addresses column: start at 128, halve down

Subnet Boundaries Chart (Optional)

Pre-written ranges to quickly locate which block an IP falls into:

- 128 addresses: 0–127, 128–255
- 64 addresses: 0–63, 64–127, 128–191, 192–255
- 32 addresses: 0–31, 32–63, 64–95, 96–127, 128–159, 160–191, 192–223, 224–255
- 16 addresses: 0–15, 16–31, 32–47, 48–63, 64–79, 80–95, 96–111, 112–127, 128–143, 144–159, 160–175, 176–191, 192–207, 208–223, 224–239, 240–255
- 8 addresses: multiples of 8 (0, 8, 16, 24...)
- 4 addresses: multiples of 4 (0, 4, 8, 12...)

Four-Step Process

1. Convert CIDR to decimal subnet mask (use chart)
2. Determine subnet/network address
3. Determine broadcast address
4. Calculate first IP (subnet + 1) and last IP (broadcast – 1)

Rules

Subnet Address

- Mask = 255 → bring down the IP address octet
- Mask = 0 → bring down 0

- Mask = interesting → find which block the IP falls into, use start of that block

Broadcast Address

- Mask = 255 → bring down the address octet
- Mask = 0 → use 255
- Mask = interesting → next block start – 1

► EXAMPLE

165.245.12.88/24

- /24 → third octet → mask 255 → full mask: 255.255.255.0
- Subnet: 165.245.12.0 (octets 1–3 mask=255 copy down, octet 4 mask=0 use 0)
- Broadcast: 165.245.12.255 (octets 1–3 copy down, octet 4 mask=0 use 255)
- First IP: 165.245.12.1
- Last IP: 165.245.12.254

► EXAMPLE

165.245.12.88/26

- /26 → fourth octet → mask 192 → full mask: 255.255.255.192
- Addresses per subnet: 64
- IP octet 4 = 88 → falls in block 64–127
- Subnet: 165.245.12.64
- Broadcast: 165.245.12.127 (next block starts at 128, minus 1)
- First IP: 165.245.12.65
- Last IP: 165.245.12.126

► EXAMPLE

165.245.12.88/20

- /20 → third octet → mask 240 → full mask: 255.255.240.0
- Addresses per subnet: 16
- IP octet 3 = 12 → falls in block 0–15
- Subnet: 165.245.0.0
- Broadcast: 165.245.15.255 (next block starts at 16, minus 1 = 15)
- First IP: 165.245.0.1
- Last IP: 165.245.15.254

► EXAMPLE

18.172.200.77/11

- /11 → second octet → mask 224 → full mask: 255.224.0.0
- Addresses per subnet: 32
- IP octet 2 = 172 → falls in block 160–191
- Subnet: 18.160.0.0
- Broadcast: 18.191.255.255 (next block starts at 192, minus 1 = 191)
- First IP: 18.160.0.1
- Last IP: 18.191.255.254

► EXAMPLE

18.172.200.77/17

- /17 → third octet → mask 128 → full mask: 255.255.128.0
- Addresses per subnet: 128
- IP octet 3 = 200 → falls in block 128–255
- Subnet: 18.172.128.0
- Broadcast: 18.172.255.255 (last address in block)
- First IP: 18.172.128.1
- Last IP: 18.172.255.254

Exam Tips

- Write chart on dry erase board at testing center or type into notepad for online exam
- Bring your own fine tip dry erase pen (check with testing center first)
- Find the shortcut that works best for you — seven second, magic number, or other

Overview

- Routers automatically discover routes and update each other
- No manual configuration of individual routes needed
- New routers added → all routers automatically learn the new routes
- Routers removed → all routers automatically remove those routes

Overhead

- Requires CPU and memory inside the router
- May need additional monitoring to ensure router can handle the load
- Requires initial configuration of the dynamic routing protocol

How Dynamic Routing Works

1. Router listens for routing updates on local subnet (sent directly or via multicast)
2. Router builds its own routing table from received updates
3. Router sends its own multicast to inform nearby routers of routes it knows
4. Receiving routers interpret updates — determine if it's a better route or secondary route
5. If network changes occur (link added/removed, router added/removed), routers inform all others

Choosing a Routing Protocol

Factors to consider:

- **Routing decisions** — based on link state (up/down) vs. hop count vs. link speed
- **Convergence time** — how quickly routers adapt to changes (seconds vs. minutes)
- **Router compatibility** — some protocols work across all manufacturers, some are vendor-specific

EIGRP (Enhanced Interior Gateway Routing Protocol)

- Cisco-centric (proprietary aspects), but may be available on other manufacturers' routers
- Relatively easy to set up
- Converges relatively quickly when changes occur
- Identifies and prevents routing loops
- Efficiently discovers other EIGRP routers with minimum network traffic

OSPF (Open Shortest Path First)

- Generic/open standard — available on many different manufacturers' devices
- Used within an Autonomous System (AS) — a network you have complete control of
- Link-state protocol — determines best route based on uptime and availability
- Assigns costs to individual links (based on throughput, link status, traversal time)
- Lowest cost and fastest path = best route
- Can load balance across links with identical costs

BGP (Border Gateway Protocol)

- External gateway protocol — routes traffic outside your AS to other organizations
- Used on WANs and internet connections
- Designed to dynamically update routes across the entire internet
- Sometimes called the "three-napkins protocol" (originally sketched on napkins)
- Used when an organization has one or more internet connections and needs dynamic routing to internet sites

Key Terms

- **Convergence** — the time it takes for all routers to update their tables after a network change
- **Autonomous System (AS)** — a network under single administrative control
- **Link-state protocol** — makes routing decisions based on link uptime and availability

The Problem

- IPv4 supports only 4.29 billion addresses — completely exhausted
- Estimated 20–30 billion devices connected to the internet
- NAT extends the functionality of IPv4

RFC 1918 Private IP Ranges

Not routable on the public internet; used internally in homes and businesses.

Range	Typical Use
10.0.0.0 – 10.255.255.255	Large enterprises
172.16.0.0 – 172.31.255.255	Mid-size networks
192.168.0.0 – 192.168.255.255	Home networks

How NAT Works (Standard)

Translates one IP address to a different IP address (private → public).

1. Internal device sends packet with private source IP
2. NAT router receives packet, recognizes private source IP
3. Router translates source IP to an available public IP from its pool
4. Packet goes to destination with public source IP
5. Return traffic comes back to the public IP
6. NAT router translates destination IP back to the original private IP
7. Internal device receives the response

Each internal device using standard NAT requires its own public IP address.

NAT Overload / PAT (Port Address Translation)

Translates both the IP address AND the port number — allows many devices to share one public IP.

How It Works

1. Internal device sends packet with private IP + source port (e.g., 10.10.20.50:3233)
2. NAT router translates to public IP + different port (e.g., 94.1.1.1:1055)
3. Router stores mapping in NAT table (private IP:port ↔ public IP:port)
4. Another device (e.g., 10.10.20.70:5782) gets the same public IP but a different port (94.1.1.1:1056)
5. Many internal devices share one public IP — differentiated by port number

► EXAMPLE

NAT Table

Private Address	Public Address
10.10.20.50:3233	94.1.1.1:1055
10.10.20.70:5782	94.1.1.1:1056

Port numbers increment for each new session — same public IP, different port.

Routing Tables

- Every device has a routing table (workstations, servers, routers)
- Router evaluates incoming traffic destination and refers to routing table to determine best route
- Troubleshooting begins and ends with the routing table

Routing Table Entry Components

```
R 10.10.30.0/24 [120/1] via 10.10.50.2, 00:00:14,
Serial0/3/1
```

Subnet ID	Metric	Next Hop	Route
Outgoing			
w/ Prefix Len			Timer
Interface			
Route Code	Administrative Distance		

- **Route code** — how the route was learned (C = directly connected, R = RIP, D = EIGRP, etc.)
- **Subnet ID with prefix length** — the network this route reaches (10.10.30.0/24)
- **Administrative distance** — trustworthiness of the route source (120)
- **Metric** — routing protocol's internal value for route preference (1)
- **Next hop** — IP address to forward traffic to (10.10.50.2)
- **Route timestamp** — how long the route has been active (00:00:14 = 14 seconds)
- **Outgoing interface** — physical interface used to reach next hop (Serial0/3/1, sometimes optional)

Route Selection: Prefix Length (Most Specific Route)

When multiple routes match a destination, the most specific prefix wins.

► EXAMPLE

Destination 192.168.1.6 with three matching routes:

1. 192.168.0.0/16
 2. 192.168.1.0/24
 3. 192.168.1.6/32 ← most specific, chosen first
- /32 = single host; longer prefix = more specific = preferred.

Route Selection: Administrative Distance

Used to break ties when identical routes come from different sources. Lower = better.

Source	Administrative Distance
Directly connected	0
Static route	1
EIGRP	90
OSPF	110
RIP	120

Route Selection: Routing Metrics

- Internal value used by a routing protocol to choose between duplicate routes within that protocol
- Cannot compare metrics across different routing protocols (BGP metrics ≠ EIGRP metrics)
- Lower metric = better route
- RIP metric = hop count
- EIGRP metric = composite calculation (different from RIP)

FHRP (First Hop Redundancy Protocol)

Problem: devices can only configure one default gateway IP address, but redundancy requires multiple routers.

Solution: create a virtual IP address (VIP) shared between routers.

How It Works

1. Active router holds the VIP — all traffic flows through it
2. Standby/backup router is on the same subnet, always communicating with the active router
3. If standby detects active router failure → standby becomes active and takes over the VIP
4. End users continue using the same default gateway — failover is seamless

Subinterfaces

- Multiple virtual interfaces assigned to a single physical interface
- Used when a trunk connection carries multiple VLANs to a router on one physical link
- Each subinterface gets its own IP address, subnet mask, and routing configuration

Naming Convention

Physical interface: ethernet1/1 Subinterfaces: ethernet1/1.10, ethernet1/1.20, ethernet1/1.100

► EXAMPLE

- Three VLANs (red, green, blue) on separate switch ports
- Single trunk cable between switch and router
- Router has subinterfaces g0/0.1, g0/0.2, g0/0.3
- Each subinterface assigned a different IP/subnet for its VLAN

How Routers Forward Traffic

1. Identify destination IP address in incoming packet
2. Examine routing table for best route
3. If destination subnet is directly connected → send to local subnet
4. If not directly connected → forward to next hop (IP of next router)
5. Next router repeats the process until packet reaches final destination
6. If no matching route found → packet is discarded

Routing Table

- Contains known routes: destination networks and their next hops
- Directly connected subnets are automatically in the table
- Remote subnets must be added (statically or dynamically)

Static Routing

Manually configuring routes in every router's routing table.

Advantages

- Quick to configure on small networks
- No overhead from dynamic routing protocols (no CPU cycles, no memory usage)
- Common for stub networks (remote sites with a single internet connection)
- Relatively secure — no dynamic routing updates that could be manipulated

Disadvantages

- Challenging on larger networks (hundreds/thousands of routers)
- Relies on administrator adding correct routes — misconfiguration can create routing loops
- Routes don't change automatically — network changes require manual updates to every affected router
- No automatic rerouting if a link goes down

Static Route Configuration Example

► EXAMPLE

Network: three routers, each with local subnets

- Router 1 directly connected: 10.10.10.0/24, 10.10.40.0/24, 10.10.50.0/24
- Router 2 directly connected: 10.10.20.0/24 (Jack's network)
- Router 3 directly connected: 10.10.30.0/24

Router 1 has no knowledge of 10.10.20.0/24 or 10.10.30.0/24 without static routes.

Adding Static Routes to Router 1

```
ip route 10.10.20.0/24 10.10.40.2
ip route 10.10.30.0/24 10.10.50.2
```

- 10.10.20.0/24 → next hop 10.10.40.2 (router 2)
- 10.10.30.0/24 → next hop 10.10.50.2 (router 3)

Key Terms

- **Next hop** — IP address of the next router that should receive the packet
- **Stub network** — remote site with a single connection; all traffic uses one path
- **Routing loop** — misconfiguration where packets cycle between routers without reaching destination

Speed and Duplex

Speed

- How fast the connection runs: 10 Mbps, 100 Mbps, 1000 Mbps (1 Gbps), 10 Gbps, or faster
- Must be the same on both sides
- Often set to auto-negotiate
- Mismatched speed → connection will not work at all (no link light)

Duplex

♦ SUPPLEMENTARY ♦

Half duplex = send OR receive, not both simultaneously. Full duplex = send AND receive at the same time.

- Half duplex or full duplex
- Must be the same on both sides
- Often set to automatic
- Mismatched duplex → connection operates but with very poor performance under load

Troubleshooting

First check speed and duplex match on both sides, then move to IP configuration.

IP Configuration

Needed for ethernet connections, VLAN configs, management interfaces, etc.

- IP address
- Subnet mask
- Default gateway
- DNS settings

Values assigned by network administrator. Incorrect configuration (wrong gateway, wrong subnet mask) → cannot connect to other IP devices.

Link Aggregation (LAG / Port Bonding)

- Connecting multiple interfaces together between two devices
- Switch interprets them as one large connection (not a loop)
- Example: four 1 Gbps links = 4 Gbps throughput between devices

LACP (Link Aggregation Control Protocol)

- Automatic configuration of link aggregation
- Configure interface as LACP, plug in cables, underlying config happens automatically

- LACP traffic visible between switches in packet captures

MTU (Maximum Transmission Unit)

- Most efficient communication avoids fragmentation
- Fragmentation slows communication; losing one fragment requires resending entire frame
- MTU usually determined automatically on first connection
- Filtering and firewalls may prevent auto-discovery → may need manual MTU configuration

Jumbo Frames

- Standard ethernet frame: 1,500 bytes
- Jumbo frame: up to 9,216 bytes MTU (many devices use 9,000 as accepted high end)
- One jumbo frame carries the same data as approximately six normal frames
- Fewer packets to switch/route → transfers more data in shorter time

Requirement

- ALL devices in the communication path must understand jumbo frames (switches, routers, endpoints)
- Any device that doesn't support jumbo frames will drop the frames → communication fails

The Loop Problem

- Two cables between two switches = loop
- No mechanism at MAC address level to count how many times a frame has been seen
- Frames loop endlessly, adding more traffic until switches are overwhelmed
- Can bring down network in seconds
- Fix: disconnect the cable creating the loop

STP (IEEE 802.1D)

Automated standard that recognizes and prevents loops on a network.

Port States

- **Blocking** — not forwarding any traffic; used to prevent loops
- **Listening** — clears existing tables, restarts the process
- **Learning** — listens to understand switch topology and detect existing loops
- **Forwarding** — traffic can be transmitted through this interface
- **Disabled** — administratively disabled; does not participate in STP

Port Roles

- **Root port (RP)** — interface on a switch connected ultimately to the root bridge
- **Designated port** — any other port that can forward traffic but is not the root port
- **Blocked port** — port disabled by STP to prevent a loop

Root Bridge

- Single switch elected as the reference point for the entire STP topology
- All other switches determine their root ports based on shortest path to the root bridge

STP Recovery (Topology Change)

1. Link failure detected (e.g., connection between switches lost)
2. STP clears existing configuration and relearns the network topology
3. Blocked ports may become forwarding; root ports may shift
4. Network recovers without creating a loop

Rapid Spanning Tree Protocol (RSTP / IEEE 802.1W)

- Original STP convergence: 30–50 seconds
- RSTP convergence: 6 seconds
- Backwards compatible with original STP
- Works very similarly to STP — just operates much faster
- Can run alongside original STP on mixed-device networks

LANs and Broadcast Domains

- A LAN is a group of devices in the same broadcast domain
- Devices connected to a switch are in the same broadcast domain
- Separate switches = separate broadcast domains (cannot communicate directly)

VLANs (Virtual Local Area Networks)

- Separate broadcast domains on the same physical switch
- Each switch interface is assigned to a VLAN
- Only devices in the same VLAN can communicate with each other
- Identified by number (VLAN 1, VLAN 2, VLAN 3, etc.)
- Reduces cost, space, power, and management — one switch instead of multiple

802.1Q Trunking

Problem: connecting VLANs across multiple switches would require one cable per VLAN.

Solution: a trunk — a single physical connection that carries traffic for all VLANs.

How Trunking Works

1. Device on VLAN 200 sends a normal ethernet frame to the switch
2. Switch adds a VLAN tag (VLAN 200) to the frame before sending it across the trunk
3. Receiving switch reads the tag, removes it, and delivers the frame to the correct VLAN

VLAN Tag

Normal ethernet frame:

Preamble	SFD	Dest	Source	Type	Payload	FCS
		MAC	MAC			

802.1Q tagged frame (VLAN tag inserted after Source MAC):

Preamble	SFD	Dest	Src	VLAN	Type	Payload	FCS
		MAC	MAC	Tag			

- 12 bits long
- Normal range: VLAN 1–1005
- Extended range: VLAN 1006–4094
- Reserved: VLAN 0 and 4095

- Total usable: 4,094 VLANs

ISL (Inter-Switch Link)

- Proprietary (Cisco) trunking protocol, predates 802.1Q
- Relatively outdated — any trunk you encounter will likely be 802.1Q

Default VLAN

- Every switch has a default VLAN (commonly VLAN 1)
- All interfaces belong to the default VLAN until reconfigured

Native VLAN

- Traffic on the native VLAN traverses a trunk WITHOUT a VLAN tag
- Used for devices that cannot communicate over 802.1Q
- Used for management traffic and switch notification messages
- Must be identical on both sides of a trunk — mismatches cause error messages in the switch log

Layer 3 Switches

- Combines layer 2 switching (MAC-based forwarding) and layer 3 routing (IP-based forwarding) in one device
- Layer 2 functionality unchanged; layer 3 functionality unchanged — just combined
- Can route between VLANs within the same switch
- Inter-VLAN routing interfaces called SVIs (Switched Virtual Interfaces)
- May need to enable layer 3 functionality (often requires switch restart)
- Routing functionality tends to be less capable than a standalone router
- Good for smaller/remote sites; larger networks still benefit from dedicated routers

Voice and Data VLANs

Problem: voice over IP and data share the same ethernet connection but have different needs.

- Data: bursty bandwidth usage
- Voice: requires consistent level of access; affected by congestion

Solution

- Put voice on one VLAN (e.g., VLAN 200) and data on another (e.g., VLAN 100)
- Single cable from switch → phone → computer

- Switch recognizes both phone and computer, uses trunking on the same cable
- Separates voice and data traffic so neither affects the other
- Requires a switch that supports recognizing both phone and computer on the same port

Wireless Mesh

- Multiple access points communicate with each other in a close area
- Client devices connect to one AP and reach the rest of the network through the mesh
- Adding new APs includes them in the mesh automatically
- Non-AP devices can also mesh in an ad hoc form
- Self-healing: if one device leaves, the mesh detects it and maintains connectivity for remaining devices

Ad Hoc (IBSS)

- Two devices connect directly using 802.11 without an access point
- Also called Independent Basic Service Set (IBSS)
- Can be long-term (two devices permanently connected) or temporary
- Common temporary use: connect to an IoT device, transfer configuration (SSID, security settings), then disable the ad hoc connection so the IoT device joins the main network

Point to Point

- Connects two locations directly (e.g., building to building)
- Each side has an AP connected to a switch and internal network
- Not all APs support point to point mode — requires both software support and hardware capable of extended distance
- May require additional antennas, power output changes, and appropriate frequencies for longer distances

Infrastructure Mode

- Most common wireless connection type (home and office)
- Centralized access point that all wireless devices connect to
- Devices communicate to the wired network or other wireless devices through the AP
- Some APs allow device-to-device wireless communication; others isolate devices to only communicate with the AP and networks beyond it

Wireless Security Goals

- Authentication: limit network access (username, password, multi-factor)
- Confidentiality: encrypt all data sent over the wireless network
- Integrity: message integrity check verifies received data matches what was sent

WEP (Wired Equivalent Privacy)

- One of the first wireless encryption methods
- Has significant cryptographic vulnerabilities
- Needed to be replaced quickly

WPA (Wi-Fi Protected Access)

- Temporary stopgap between insecure WEP and a more secure replacement
- Designed to work on the same hardware as WEP
- Known to need enhanced security beyond what this initial version provided

WPA2 (Wi-Fi Protected Access Version 2)

- Available since 2004
- Uses CCMP (Counter Mode with Cipher Block Chaining Message Authentication Code Protocol / Counter/CBC-MAC)
- Combines encryption and integrity in the same protocol
- Encryption: AES
- Message integrity check: CBC-MAC

WPA3 (Wi-Fi Protected Access Version 3)

- Introduced in 2018
- Uses GCMP (Galois Counter Mode Protocol)
- Encryption: AES
- Message integrity check: GMAC (Galois Message Authentication Code)

Best Practice

- Use the highest level of security available on APs and client devices
- Upgrade all devices to use the most capable security method available

Service Sets

Independent Basic Service Set (IBSS)

- Direct device-to-device communication without an access point
- Also called an ad hoc connection
- Common use: configuring IoT devices (door locks, lights) before connecting them to the main network

SSID (Service Set Identifier)

- The name of the wireless network
- Appears in the list of available networks on client devices
- Can be configured on multiple access points for coverage across large areas

BSSID (Basic Service Set Identifier)

- The hardware address of an individual access point
- Differentiates between access points that share the same SSID
- Example format: 60:3D:26:11:22:33

ESSID (Extended Service Set Identifier)

- The shared network name used across multiple access points
- Enables seamless roaming — device moves between APs without notification or interruption
- All APs share the same SSID but have different BSSIDs

Captive Portals

- Screen presented when first connecting to a wireless network
- May require agreement to terms or authentication (username, password, or other factor)
- Centralized access table tracks authenticated devices
- Unauthenticated devices are redirected to the captive portal
- Access is granted for a predefined time (e.g., 24 hours), then reauthentication is required

Wireless Security Modes

Mode	Description
Open	No security, no authentication, anyone can communicate
OWE	Encrypts traffic but prevents device-to-device communication
WEP	Legacy encryption (insecure)
WPA/WPA2/WPA3	Progressively stronger wireless security standards
Personal (PSK)	Pre-shared key — everyone uses the same password
Enterprise (802.1X)	Individual username/password; disabling account removes access

- PSK example: coffee shop posts the shared password for all customers
- Enterprise: users authenticate with their own corporate credentials

Antenna Types

Omnidirectional

- Distributes signal evenly in all directions
- Common on consumer/home access points
- Ideal when AP is centrally located
- Wasteful if AP is in a corner — signal goes where no one connects

Directional

- Focuses signal in a single direction
- Useful for corner placement or building-to-building links
- Measured in decibels of gain
- Every 3 dB of gain doubles the power

Directional Antenna Types

Type	Characteristics
Yagi	Very directional, high gain
Parabolic	Focuses signals into a single feed horn; useful over long distances

Access Point Management

Autonomous Access Points

- Standalone devices that operate independently
- No additional hardware or software required
- Common in home environments

Lightweight Access Points

- Hardware mounted in ceiling; configuration/intelligence maintained on the switch
- Less expensive to deploy

- Managed centrally

CAPWAP (Control and Provisioning of Wireless Access Points)

- Standard for managing wireless access points from one central station
- Central station = wireless LAN controller (WLC)
- Provides a “single pane of glass” view of the wireless infrastructure

Wireless LAN Controller Functions

- Deploy new access points
- Monitor existing AP performance
- Push configuration changes to some or all APs
- Generate usage reports (how much bandwidth used, by whom)
- Usually paired with APs from the same manufacturer

IEEE 802.11 Standards

- IEEE 802.11 committee creates worldwide standards for wireless networking
- Standards have marketing-friendly names:

Standard	Wi-Fi Name
802.11ac	Wi-Fi 5
802.11ax	Wi-Fi 6/6E
802.11be	Wi-Fi 7

- Future major releases will increment the Wi-Fi number (Wi-Fi 8, 9, etc.)

Wireless Frequencies and Channels

- 802.11 uses three frequency ranges: 2.4 GHz, 5 GHz, and 6 GHz
- Access points may support one or more of these simultaneously
- Channels are named groupings of frequencies for easy reference
 - Channel 6 = centered on 2.437 GHz
 - Channel 44 = centered on 5.220 GHz

Channel Bandwidth

- Each channel uses a range of frequencies around its center (the bandwidth)
- Common bandwidths: 20 MHz, 40 MHz, 80 MHz, 160 MHz

Frequency Comparison

Band	Channels Available	Typical Bandwidth
2.4 GHz	3 nonoverlapping	20 MHz
5 GHz	Many more	20–160 MHz
6 GHz	Even more	20–160 MHz

- More available channels = less conflict with nearby networks

Band Steering

- Modern devices can support multiple frequency bands
- Without band steering: device connects to whichever frequency has the strongest signal
- Strongest signal does not always mean best throughput
- Band steering lets the AP administrator control which frequency a dual-band client uses
 - Example: steer 2.4/5 GHz capable devices to prefer 5 GHz

Regulatory Impacts

Country-Specific Regulation

- Each country manages its own frequency allocations

- US: Federal Communications Commission (FCC)
- Countries work together to coordinate frequency use

802.11h Interoperability

- Adds features so multiple wireless networks can coexist in one area
- Follows guidelines from the International Telecommunications Union (ITU)
- Two key features: Dynamic Frequency Selection (DFS) and Transmit Power Control (TPC)

Dynamic Frequency Selection (DFS)

- APs automatically select channels that won't conflict with existing networks
- If the wireless environment changes, the AP changes channels and tells all connected clients to move with it
- Eliminates need for manual channel configuration

Transmit Power Control (TPC)

- AP controls client transmit power (rather than client setting its own)
- AP can tell a client to reduce power
- Allows many devices to operate at optimal power without conflicting with other services

Data Center Power Usage

- Approximately 9% of all US power consumption is used by data centers
- Much of this power goes to managing the environment (humidity, temperature, other controls)

Humidity

- High humidity → condensation (water in data center)
- Low humidity → excessive static discharge (harmful to equipment)
- Ideal range: 40%–60%
 - High enough to prevent static discharge
 - Low enough to prevent condensation
- Can vary by geography

Temperature

- Equipment generates heat that must be cooled
- Optimal range: 64–81°F
- Cooling equipment constantly works to maintain this range
- External influences can affect temperature:
 - Outdoor temperature increases/decreases add load to cooling systems
 - Increased system load creates more heat, requiring additional cooling

HVAC

- Well-engineered system designed to maintain specific humidity and temperature
- Sensors located throughout the data center
- Ensures consistent temperature and humidity across the entire facility

Fire Suppression

- Data centers contain large amounts of electrical equipment — water is not suitable
- Tank in back of data center contains inert gas or chemical agent
- Designed to suppress fire or remove oxygen from the air
- When fire alarm is pulled, chemicals are dispersed throughout the entire data center
- May cause damage to equipment but prevents fire from spreading
- Integrated with HVAC: fire detection shuts down HVAC to prevent oxygen from feeding the fire

Distribution Frames

♦ SUPPLEMENTARY ♦

Terminating = connecting the end of a cable to a connector or contact point (RJ45 plug, punch down block, fiber connector, etc.) so it can carry a signal and interface with equipment.

Punch down = a termination method where wires are pressed into slots on a punch down block using a special tool. The tool forces wire into a metal contact that cuts through insulation and makes the electrical connection — no stripping or soldering needed.

- Area where cables are passively terminated (punch down blocks, patch panels)
- Usually on the back wall of a data center
- Terminates LAN and voice communication cables
- The entire room is often referred to as the distribution frame room

Main Distribution Frame (MDF)

- Single room — the data center or central point of the network
- WAN connections and internal LAN connections are punched down here
- Perfect testing point for both internal and external connections
- Even without large punch down blocks, the central data center room is often called the MDF

Intermediate Distribution Frame (IDF)

- Usually on a separate floor or in a separate building
- Directly connected to the MDF
- Includes switches, routers, and other equipment
- Common in medium to large environments
- Users on a floor connect to a switch in the IDF, which connects back to the core switch in the MDF

Racks

- Standard width: 19 inches
- Equipment designed to fit the 19-inch width
- Height measured in rack units (U) — 1U = 1.75 inches
- Most racks are approximately 42U in height
- Depth varies — must match equipment length
- Equipment can be stacked to maximize space

Locked Racks

- Provide additional security and control over components
- Installed side by side to optimize data center space

- Door on front with lock to prevent unauthorized access
- Ventilation in front, top, and bottom for proper cooling

HVAC (Heating, Ventilating, and Air Conditioning)

- Required to cool data center equipment running constantly
- Must support the heat load, have sufficient power, and integrate with fire systems
- Much more complex than a simple air conditioner

Hot and Cold Aisles

- Cold aisle: front of servers — cold air pulled in
- Hot aisle: back of servers — hot air blown out
- Equipment must be installed with correct orientation (intake from cold aisle, exhaust to hot aisle)

Airflow Cycle (Raised Floor)

1. HVAC creates cold air sent under a raised floor
2. Vents in floor allow cold air into the cold aisle
3. Servers pull in cold air from the front
4. Heated air exits the back into the hot aisle
5. Hot air rises to the ceiling
6. HVAC pulls hot air back in, re-cools, and repeats

Alternative (No Raised Floor)

- Cold air blown directly into cold aisle from above
- Cold aisles may be covered with plastic to contain cold air
- Hot air goes back into the ceiling and returns to HVAC

Cable Infrastructure

Patch Panels

- Single cable runs from each desk back to the IDF
- Back side: 110 block where wires are punched down
- Front side: RJ45 modular connectors
- Patch cables connect from patch panel RJ45 to switch ports
- Installed cables from patch panel to desks are never moved once installed
- Numbered connections identify which desk each port serves

Moves, Adds, and Changes

- Move: relocate a patch cable to a different connection
- Add: connect new employees without running new cables (already run to desks)
- Change: only internal IDF connections need to change

Fiber Optic Distribution Panel

- Brings in fiber runs from other buildings or floors

- Must not exceed the bend radius when routing fiber
- Large loops maintain proper bend radius inside the panel
- Service loop: extra fiber wrapped around for future flexibility (extend or move without rerunning fiber)

Electrical Safety

- Never put yourself in contact with any voltage
- Always disconnect from power source before working on a device
- Some devices store charge in capacitors (CRT monitors, laser printers) — can shock even when disconnected
- Never connect any part of your body to a ground wire or anything that might be energized

Electrical Terms

Term	Abbreviation	Description	Hose Analogy
Ampere	A (amp)	Rate of electrons moving past a point per second	Hose diameter
Voltage	V (volt)	Pressure of electricity flowing through a wire	Water pressure
Watt	W	Power being used — calculated as volts × amps	Total water flow

- Example: $120V \times 0.5A = 60W$

AC (Alternating Current)

- Current changes direction as it moves through wire (represented by wavy line)
- Easy to distribute over long distances — used for homes and offices
- US/Canada: 110–120V at 60 Hz
- Europe: 220–240V at 50 Hz

DC (Direct Current)

- Current moves in a single direction with constant voltage
- Symbol: solid line on top with dashed lines below
- Electrical components use DC to operate
- Power supplies in devices (or in the power cord) convert AC to DC

UPS (Uninterruptible Power Supply)

- Connected to power source with internal batteries for backup
- Protects against:
 - Power outages
 - Brownouts (voltage drops)
 - Surges (voltage spikes)

UPS Types

Type	Description
Offline/Standby	Activates after power loss; small gap may reboot devices
Line-Interactive	Compensates for brownouts by increasing power when voltage drops
Online/Double-Conversion	Always runs from battery; losing power has no effect on systems

UPS Features

- Automatic shutdown when batteries are low
- Varying battery capacities
- Different numbers of outlets
- Additional interfaces for phone lines or cable modem connections

PDU (Power Distribution Unit)

- Looks like a surge suppressor but is an intelligent device
- Rack-mountable
- Allows remote connection and management of each interface
- Can power off/on devices from a remote location
- Has an ethernet connection with an IP address and built-in web server

Need for Configuration Management

- Changes are inevitable: OS updates, patches, application updates, firewall configs, new applications
- Need a formal process — know when changes occur, plan for them, inform everyone
- Documentation is critical for rebuilding systems after a disaster

Production Configuration

- The configuration used when a switch, router, or firewall is working perfectly after initial setup
- Standard configuration rolled out for each production system
- Covers all aspects: hardware/firmware versions, device driver updates, software versions
- Certain software versions may work better or worse than others
- Extensive testing done behind the scenes before deploying to production
- Not installed directly into production to “see how it runs”

Backup Configurations

- Not every scenario can be tested in lab — need a plan to revert if something goes wrong
- Standard operating procedure: make a backup before any change
- Applies to firewalls, switches, routers, operating systems, and anything else being changed
- Methods:
 - Copy files that will be updated (e.g., a single configuration file)
 - VM snapshot — saves files, configuration, and everything at that point in time
- Can revert to backup immediately if there’s a problem
- Also useful for reverting later if you decide to go back to previous configuration

Baseline / Golden Configuration

- Deploying an application involves many components (workstation configs, firewall changes, application server)
- Document every aspect of the application installation
- Golden configuration certifies the application will work properly if all configs are in place
- Used to verify proper software and correct configurations

- Integrity checks compare production configuration against the baseline/golden config
- If differences found:
 - Update production to match golden config, OR
 - Update baseline to match new production configuration
- After changes, the new version becomes the baseline for integrity checks going forward

End of Life (EOL)

- Manufacturer no longer supports a product
- May continue providing security patches and updates
- No new software versions or feature enhancements
- Triggers planning and budgeting for replacement over time

End of Support (EOS)

- No patches, updates, new features, or anything from the manufacturer
- Significant security concern — no future security updates ever
- More urgent than EOL

Patches and Bug Fixes

- Important for maintaining uptime, availability, and stability
- Also close security holes
- Many organizations provide monthly updates on a recurring schedule
- Allows planning deployment around known time frames
- Occasional out-of-schedule updates for zero-day or significant security events

Operating System Updates

- Constant updates across Windows, Linux, iOS, Android, etc.
- Include bug fixes and security patches
- Also internal configuration changes:
 - User account modifications
 - Minimum password length increases
 - Password complexity changes
 - Access method modifications
 - Built-in firewall configuration changes (limit by IP/URL)
 - Anti-malware/antivirus updates

Firmware Updates

- Software running on purpose-built appliances/hardware (printers, cable modems, etc.)
- Devices have their own embedded operating system — no direct OS access
- Updates close security holes
- May be done over the network or require physical connection

- Best practice: save firmware binaries for rollback if new version has problems
- Challenge: hardware manufacturers may not maintain firmware promptly
 - Example: Trane Comfortlink II thermostat — vulnerabilities reported April 2014, final fix January 2016

Decommissioning

- Process of disposing of an asset no longer needed
- Applies to desktops, laptops, mobile devices, and other outdated equipment
- Must sanitize media or destroy the device to protect data
- May involve legal issues — certain data cannot be destroyed
- May need to store equipment securely until proper disposal is possible
- Never dispose of in normal trash — someone will access the data
- Recycling must protect confidentiality of data on devices

Change Management

- Managed process for tracking, monitoring, and modifying changes
- Covers: software upgrades, firewall configurations, router tables, other modifications
- Clearly defined process includes:
 - How often changes can be made
 - Window for making changes
 - Process for installing changes
 - Process for rolling back if problems occur
- Everyone must understand and follow the centralized change management process

Service Requests / Help Desk

- Help desk receives phone calls and inputs tickets into a process tracking system
- Tickets are triaged and assigned to someone who can resolve the problem
- Ticket is closed after resolution, then move to next
- Almost every organization has a process tracking system

Network Maps

Physical Network Map

- Shows physical layout of equipment, cabling, and connections
- Can track cables back to equipment in the room
- Example: internet → router (10.1.1.1) → firewall → core router (10.1.10.1) with physical wires between each

Logical Network Map

- Higher-level view of connectivity — not focused on individual interfaces/cables
- Shows how the entire network is connected (e.g., headquarters connected via a cloud)
- Useful for planning additional locations

Tools

- Visio, OmniGraffle, Gliffy.com, and others
- Can create both physical and logical maps

Rack Diagrams

- Physical view as if standing in front of the rack
- Useful when access to data center is restricted
- Allows remote guidance (e.g., “go to unit 14 for the power strip”)
- Can include rack location (row and rack identifier)
- Allows specific instructions like “reboot the server at unit 15”

Cable Maps

- Physical diagram of where wires are installed in a facility
- Usually posted on the wall in the IDF or MDF
- Shows wires running under floor or above ceiling
- Each drop is numbered — correlates desk drops with patch panel connections
- Useful during installation and troubleshooting

Layer 1/2/3 Network Diagrams

- Combine physical, data link, and network layer views
- Layer 1: physical interfaces where wires connect
- Layer 2: MAC addresses for all components
- Layer 3: IP addresses associated with each MAC address
- Shows which IP → MAC → physical interface

Asset Management

Asset Tags

- Labels on every asset (laptops, desktops, routers, firewalls, switches)
- Used to reference devices in trouble tickets
- Used for financial tracking (depreciation, purchase date, warranty status)
- Visible so support can easily locate them
- May include barcode or RFID
- Identifies owner if a third party finds the device

Asset Database

- Centralized database combining: asset tag, device components, and assigned user
- Used by support, accounting, and finance
- Tracks physical location (via assigned user’s location)
- Tracks warranty status
- Tracks software/licenses — determines how many licenses to purchase

IP Address Management (IPAM)

- Solution to plan, track, and configure DHCP and IP address schemes
- Documents which users are using which IP addresses
- Maps users to IP addresses at specific dates/times (useful with dynamic addressing)
- Identifies shortages or problems with IP address configurations
- May lead to modifying IP ranges or adding addresses to DHCP
- Manages all IPv4 and IPv6 addressing from one central console

Service Level Agreement (SLA)

- Contract defining minimum level of service from a provider
- Example: 99.99% uptime requirement
- Example: no more than four hours of unscheduled downtime
- Provider may quickly dispatch technicians or keep spare equipment on-site

Site Surveys

- Identify all access points in use (including ones not under your control)
- Document frequencies in use

- Configure systems for minimum interference
- Important in multi-company buildings or large campuses
- Should be performed often in changing environments
- Heat map: walk around with mobile device to document wireless signal propagation

NetFlow

- Summary of statistics based on traffic flows traversing the network
- Standard collection method; many products from many manufacturers
- Components:
 - **Probe:** sits on the network via physical tap or SPAN port to capture raw packets and compile statistics
 - **Collector:** central destination that receives summaries from all probes; stores data over time
 - **Report generator:** often built into the collector; provides real-time views and long-term scheduled reports
- Data includes: top conversations (ingress/egress), top endpoints, traffic by port/application, events

Protocol Analyzer

- Captures and displays raw packets (exact bits and bytes) sent across the network
- Works on wired and wireless networks
- Presents data in human-readable summary with drill-down into packet details
- Use cases: finding unknown traffic, troubleshooting slow applications
- Some organizations store all captured packets long-term on large storage arrays for later analysis

Network Performance Baseline

- Picture of what a normal day on the network looks like, built from collected data
- Useful for troubleshooting — compare current values to what's normal
- Can drill down to individual device performance
- Management console or SIEM can identify anomalies by comparing against the baseline

Syslog

- Standard protocol for transferring log files from network devices to a central location
- Devices (switches, routers, firewalls, servers) each configured with syslog destination
- Commonly sent to a SIEM
- Log entries include:
 - Device identifier (which device sent it)
 - **Facility code:** the program that created the log entry
 - **Severity level:** importance of that log entry

- Sensitivity settings can be adjusted to only capture warning/urgent info and exclude informational details

SIEM (Security Information and Event Manager)

- Central system collecting log files from many devices
- Stores data on large drive arrays
- Capabilities:
 - Real-time dashboard with alerts and alarms (e.g., failed authentications, devices going offline)
 - Advanced reporting — compare and contrast statistics across multiple devices over time
 - Ad hoc queries (e.g., search for "fail*" AND "password" to find authentication failures)
 - Forensics — go back in time to track authentication events and service usage
- Can track traffic flow across the entire path: server → switch → firewall → internet

API Integration

- Central management station communicates to devices via Application Programming Interface
- Eliminates need to SSH into each device individually
- Enables scripts and programs to send commands, make configuration changes, or query information
- Essential for large environments (hundreds or thousands of devices)
- Changes can be applied to many devices automatically with one action

Port Mirroring / SPAN

- Redirects a copy of traffic from one switch port to another port connected to a monitoring device
- Also called Switched Port Analyzer (SPAN)
- Use cases: IDS/IPS, performance monitoring, NetFlow collection
- Some switches support mirroring traffic from one switch to another (remote SPAN)
- Can mirror single interfaces, multiple interfaces, or entire VLANs
- Original traffic is unaffected — only a copy is sent to the monitoring device

Network Tap

- Physical device connected between two devices to capture traffic
- Alternative to port mirroring for gathering raw packets

Network Discovery

- Techniques for identifying devices on the network:
 - **LLDP** (Link Layer Discovery Protocol): standard discovery protocol on switches
 - **CDP** (Cisco Discovery Protocol): Cisco's proprietary version
 - Ping scans and port scans
 - Commercial network scanners
 - SNMP
- Some organizations run discovery scans nightly to track what's installed
- Others run scans only during troubleshooting
- Daily scans + reporting can reveal what traffic is on the network and which devices are sending it

Traffic Analysis

- Detailed frame-by-frame / packet-by-packet description of traffic flowing across the network
- Often summarized into simplified views
- Firewall log data includes:
 - Timestamp of each traffic flow
 - Protocol (TCP/UDP)
 - Port numbers
 - Client IP address and server IP address
 - Host names (via DNS resolution)
 - Bytes transferred
- Stored long-term for reports and forensics

Performance Monitoring

- Gathering overall performance data from devices:
 - Network utilization statistics
 - Error overview
- Methods: SNMP, NetFlow statistics, protocol analysis (frame level), software agents on devices
- Helps determine if network is highly utilized or experiencing errors

Availability Monitoring

- Simple metric: device is up (green) or down (red)
- Provides instant visibility into device status
- Can trigger alarms and alerts:
 - Email messages
 - Helpdesk tickets

- Notifications to responsible staff
- Real-time view of current status
- Historical data collected over time for availability reports
- For deeper performance details, drill down into NetFlow or SNMP

Configuration File Management

- Devices (switches, routers, firewalls) have configuration files
- Can back up and restore configurations
- Configuration files may be version-specific:
 - Older config may not load on a newer software version
 - May need to downgrade firmware/software to restore an older config
- Must store both configuration files and previous firmware/software versions
- Multiple identical devices (e.g., 10 web servers): compare and contrast configs to verify consistency
- Ongoing monitoring of config files with alerts if unauthorized changes occur
- Integrated with change control process — standardized method for modifying configurations

Overview

- Standard protocol for monitoring and configuring network devices
- Provides a management interface regardless of device manufacturer
- Central network management console queries devices via SNMP
- Queries request specific data (e.g., bytes into a particular interface)
- Response values stored on the management station

MIB (Management Information Base)

- Central database of parameters stored on each managed device
- SNMP queries specific values using an Object Identifier (OID)
- OID is a series of numbers representing a hierarchical path (e.g., 1.3.6.1.2.1.11.28.0)
 - 1 = ISO, 3 = org, 6 = DoD, 1 = internet, 2 = management, etc.
- MIB2 (SNMPV2-MIB): standardized set of OIDs identical across devices
- Manufacturer-specific OIDs: unique variables for a specific device
 - Manufacturer provides documentation or a MIB file to import into the management station
- MIB browser/walker: software that cycles through every possible MIB value on a device

Polling

- All SNMP polling occurs over **UDP port 161**
- Management station queries devices at regular intervals (every 1–5 minutes)
- Collected data over time builds performance graphs and visualizations
- Limitation: problems not detected until next polling interval

SNMP Traps

- Proactive alarm sent from device to management station without polling
- Uses **UDP port 162**
- Configured on the device with threshold conditions (e.g., CRC error count increases by 5)
- Management station receives trap and can alert staff or run remediation scripts

SNMP Versions

Version	Encryption	Bulk Queries	Authentication
v1	None	No	Community string
v2c	None	Yes	Community string
v3	Yes	Yes	Username + password hash

Version 1

- Original version — queries structured MIB tables
- All data sent in the clear (no encryption)

Version 2c

- Improved efficiency — can query large chunks of data at once
- Still no encryption

Version 3

- Adds encryption, message integrity, and authentication
- Most secure — recommended for production use

Authentication

Community Strings (v1/v2c)

- Simple password for accessing SNMP data
- Multiple strings per device:
 - **Read-only** (default: `public`)
 - **Read-write** (default: `private`)
 - **Trap** community string
- Relatively simplistic security

SNMP v3 Authentication

- Username and password sent as a password hash
- Much more secure than community strings

Disaster Recovery Plan (DRP)

- Covers every aspect and detail of handling outages or significant problems affecting organizational goals
- Technologies involved in recovery:
 - Backups
 - Offsite data replication
 - Cloud-based alternatives (recreate on-site servers in the cloud)
 - Completely separate remote site
- Third-party services:
 - Contract for a temporary facility to move operations to
 - Recovery services that come in and manage the disaster recovery process

Recovery Metrics

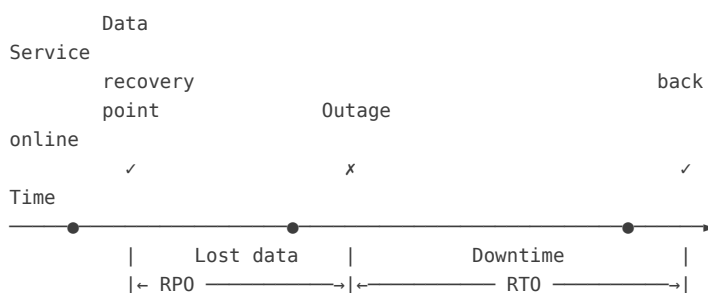
Recovery Time Objective (RTO)

- Amount of time to get back up and running after an outage
- Goal: as close to zero as possible
- Measured from the outage to when normal service level is restored
- Example: web server fails, RTO = 1 hour to become available again

Recovery Point Objective (RPO)

- Amount of data (measured in time) lost when an outage occurs
- Goal: as close to zero as possible
- Represents the gap between the last data recovery point (backup/replication) and the outage
- Determined based on available resources and backup frequency
- Varies by data type:
 - Banking transactions / patient info: very short RPO (< 1 hour)
 - Website updates / internal documents: longer RPO (1–2 hours)

RTO vs RPO on a Timeline



- Data recovery point (backup/replication) occurs

- Outage happens sometime after
- **RPO** = time between the data recovery point and the outage (data lost)
- **RTO** = time between the outage and services coming back online (downtime)

Mean Time to Repair (MTTR)

- Average time to resolve an issue (statistical, based on historical data)
- Example: average time to replace a failed router

♦ SUPPLEMENTARY ♦

MTTR vs RTO: RTO is a forward-looking target (how long recovery *should* take). MTTR is a backward-looking average (how long repairs *actually* take across multiple incidents).

Mean Time Between Failures (MTBF)

- Average time a device is expected to operate before failing
- Presented as a single time frame based on multiple criteria
- Example: firewall with MTBF of 20 years
- Helps plan how many backup units to purchase

Site Resiliency

- Process of moving operations from primary data center to a temporary facility and back
- Steps:
 1. Prepare disaster-recovery site (power, hardware, data)
 2. Move from primary location to backup facility when disaster occurs
 3. Work from backup facility until problem is resolved (hours to months)
 4. Move assets and data back to original location

Cold Site

- Empty building with no equipment, no data, no people
- Must bring backup tapes/equipment and transport people
- Lot of work when disaster is called
- Least expensive option

Hot Site

- Exact replica of primary data center
- Same hardware (purchase additional units when buying for primary)
- Applications and data replicated/backed up continuously
- Can move in and be up and running relatively quickly
- No need to install hardware, applications, or recover from backups

- Most expensive option

Warm Site

- Between a cold site and a hot site
- Some level of infrastructure: power, racks, possibly some hardware
- Need to bring data and recover from backup tapes
- Different levels of service available — choose how much hardware/data to stage

Disaster Recovery Testing

Tabletop Exercise

- Everyone sits around a conference table and steps through simulated disaster scenarios
- Describes what each person/department would do
- No physical movement of backups or equipment
- Verifies logistics are in place
- Takes about a day or two
- All key players participate

Validation Tests

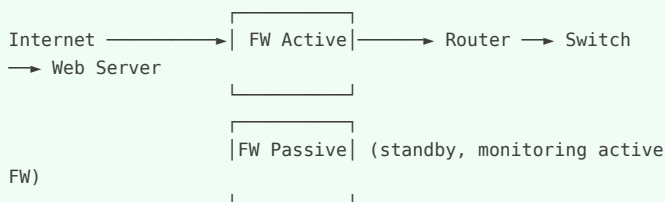
- Full-blown disaster-recovery site test (once or multiple times per year)
- Follow a specific scenario (e.g., fire destroys building, geographic evacuation)
- Go through the same process as actual disaster recovery without moving production
- Document what worked and what needs to be fixed
- Allows ongoing improvements for efficiency

Active-Passive

- Two separate pieces of equipment, only one active at any time
- Devices constantly communicate status between each other
- If primary fails, secondary takes over as the primary device
- Configuration between devices must be identical
- Changes to primary config must be copied to secondary
- Real-time information (session tables, routing tables) must also be maintained on secondary
- Secondary has exactly the same configuration as primary so it can take over seamlessly

► EXAMPLE

Firewall Active-Passive



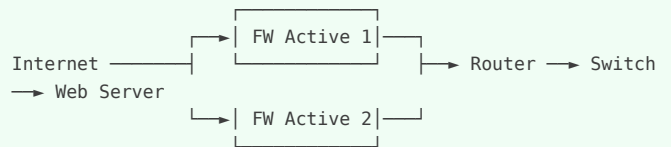
- Traffic passes through the active firewall
- If active firewall fails (power supply, software crash), passive detects it is offline
- Passive firewall promotes itself to active and becomes the primary device
- Future communication uses the new active firewall

Active-Active

- Both devices operating simultaneously
- Not as simple as turning both on — requires additional engineering
- Must track data flows across both devices (one direction through one device, return through the other)
- Requires understanding of:
 - Traffic flow directions
 - Routing configuration
 - Switch locations
 - Expected normal traffic flow patterns
- If one device fails, the remaining device continues handling all traffic — no failover process needed

► EXAMPLE

Firewall Active-Active



- Traffic flows split across both firewalls
- One flow may go through FW1, another through FW2
- If one fails, remaining firewall handles all load
- Traffic continues to flow normally

DHCP Scope

- A scope defines the pool of addresses and settings a DHCP server hands out
- Scope contents:
 - IP address range (contiguous pool, e.g., .1 through .100)
 - Excluded addresses (addresses within the range that should not be assigned)
 - Subnet mask
 - Lease duration
 - Optional settings (DNS servers, default gateway, VoIP server, etc.)
- One scope per subnet — each scope is responsible for its subnet's addresses

Address Assignment

- When a device requests an address, the server picks one from the pool and assigns it
- A lease period is associated with the address — the device can only use it for a limited time
- If the lease is not renewed, the address returns to the pool and can be assigned to another device
- The server tracks MAC-to-IP pairings
- If a device returns to the network and its previous IP is still available, the server reassigns the same IP

Address Reservation

- Assigns a fixed IP to a specific MAC address via the DHCP server
- Use case: printers, servers, or any device that needs a consistent IP
- Advantage over static configuration: manage all addresses centrally; changes don't require visiting each device
- Also called: static DHCP assignment, static DHCP, or IP reservation
- Reservation table contains: MAC address, reserved IP address, device name

DHCP Lease

- IP addresses are temporary — leased for a configured duration
- When a device shuts down, it can release the IP back to the pool
- Restarting or reconnecting reinitializes the DHCP process and may restart the lease

Lease Renewal Timers

Timer	Default	Behavior
T1	50% of lease time	Device attempts to renew with the original DHCP server
T2	87.5% (7/8) of lease time	Device attempts to renew with any available DHCP server

► EXAMPLE

Lease Timeline (8-day lease)

1. Day 0–4: Normal operation, no renewal needed
2. Day 4 (T1): Renewal period begins — device contacts original DHCP server
 - If successful: lease resets, timer restarts
3. Day 7 (T2): Rebinding period — if original server unavailable, device contacts any DHCP server
 - If successful: lease resets with new server

DHCP Options

- Additional TCP/IP settings delivered alongside IP/subnet/gateway
- Defined in the DHCP RFC (originally called vendor extensions in BOOTP)
- 254 usable option numbers available
- Not all DHCP servers support all options — verify server capabilities
- Common options:
 - IP address, subnet mask, default gateway, DNS servers (standard)
 - Option 129: Call server IP address (VoIP)
 - Option 135: HTTP proxy
- Configured under Server Options in the DHCP scope; delivered to all clients connecting to that scope

Background

- Previously all IPv4 settings (IP address, subnet mask, DNS, etc.) were configured manually on every device
- **BOOTP** (Bootstrap Protocol) was created to automate this, but had limitations:
 - Didn't configure everything automatically
 - Could not recognize when an IP address became available after a device left the network
- **DHCP** (Dynamic Host Configuration Protocol) replaced BOOTP with full automatic address configuration

DORA Process

Four steps to obtain an IP address automatically:

Discover

- Client sends DHCP Discover packet
- Source IP: 0.0.0.0 (client has no IP yet)
- Destination IP: 255.255.255.255 (broadcast)
- Source port: UDP 68
- Destination port: UDP 67
- Broadcast so all devices on subnet see it — DHCP servers can respond

Offer

- DHCP server responds with a DHCP Offer
- Source IP: DHCP server's IP (e.g., 10.10.10.99)
- Destination IP: 255.255.255.255 (broadcast, since client has no IP)
- Source port: UDP 67
- Destination port: UDP 68
- If multiple DHCP servers exist, client may receive multiple offers

Request

- Client chooses one offer and sends a DHCP Request
- Source IP: 0.0.0.0 (still no IP)
- Destination IP: 255.255.255.255 (broadcast)
- Source port: UDP 68
- Destination port: UDP 67

Acknowledge

- DHCP server sends DHCP Acknowledgment
- Source IP: DHCP server's IP
- Destination IP: 255.255.255.255 (broadcast)
- Source port: UDP 67
- Destination port: UDP 68

- Client can now configure all IP settings automatically

DHCP Ports

Direction	Port
Client (source)	UDP 68
Server (source)	UDP 67

DHCP Limitation: Broadcasts Don't Cross Routers

- All DHCP messages are broadcasts (255.255.255.255)
- Broadcasts only reach the local subnet — they do not traverse routers
- Problem for enterprise networks wanting redundant/centralized DHCP servers

DHCP Relay (DHCP Helper)

- Router feature that forwards DHCP traffic to a server on a different subnet
- Configured on the router with the DHCP server's IP address
- Process:
 1. Client broadcasts DHCP Discover on local subnet
 2. Router with DHCP relay receives the broadcast
 3. Router changes source IP to its own interface IP and destination to the DHCP server IP (broadcast → unicast)
 4. DHCP server responds with unicast back to the router
 5. Router converts the response back to a broadcast on the client's local subnet
 6. Repeats for all four DORA phases
- Allows clients to receive IP addresses from a DHCP server on a completely different subnet

Resource Records

- Individual entries in a DNS configuration file
- Each record provides different types of information:
 - Name-to-IP resolution
 - Certificates for DNS validation
 - Aliases of host names
 - Mail server locations
 - And many others

SOA (Start of Authority)

- First record at the top of a DNS configuration file
- Provides an overview of the DNS zone:
 - DNS zone details (which domain this config belongs to)
 - Serial number
 - Retry information
 - Expiration details
 - How long information should be stored

A and AAAA Records

- Define the IP address of a host — the core record for name resolution
- A record: IPv4 address
- AAAA record: IPv6 address
- Same functionality, different IP version
- Example A record: `www.professormesser.com IN A 162.159.246.164`

CNAME (Canonical Name) Record

- Creates an alias that points to another name
- Use case: single server with one IP address referenced by multiple names
- Querying a CNAME requires two lookups:
 1. DNS returns the canonical name (the target)
 2. If the target IP is not cached, a second query resolves the target's IP
- Example:
 - `chat.example.com → CNAME → mail.example.com`
 - `ftp.example.com → CNAME → mail.example.com`
 - `www.example.com → CNAME → mail.example.com`

MX (Mail Exchanger) Record

- Specifies the mail server responsible for receiving email for the domain

- Resolving the MX record gives a name (e.g., `mail.example.com`), not an IP
- A separate A record lookup is needed to get the mail server's IP address

TXT (Text) Record

- Stores human-readable text information in DNS
- Used for multiple purposes:

SPF (Sender Policy Framework)

- Specifies which email servers are authorized to send mail on behalf of the domain
- Prevents unauthorized servers from spoofing email
- Receiving mail servers check the SPF record to verify the sender is authorized

DKIM (Domain Keys Identified Mail)

- Digitally signs outgoing emails
- The public key needed to verify signatures is stored in a TXT record
- Receiving servers use the public key from DNS to verify the digital signature

NS (Name Server) Record

- Specifies where the name servers for the domain are located
- Critical for name resolution — without NS records, the domain cannot be found
- Multiple NS records provide redundancy (e.g., `ns1.example.com`, `ns2.example.com`)

PTR (Pointer) Record

- Used for reverse DNS lookups — resolves an IP address to a name
- Separate from forward (A/AAAA) records; must be configured independently
- IP addresses are stored in reverse order in the DNS configuration
- Example: querying `192.168.23.15` returns `www.example.com`

DHCPv6

- DHCP for IPv6 uses a similar process to IPv4 DHCP
- Uses redundant DHCP servers for enterprise readiness
- Managed by the DHCP administrator just like IPv4

Stateless Addressing

- Unlike IPv4, IPv6 devices can assign an IP address to themselves and communicate on the network
- No DHCP server manages the process
- No need to track IP addresses or MAC addresses
- No lease time — the address never needs to be given up

Neighbor Discovery Protocol (NDP)

- Replaces ARP from IPv4
- ARP uses broadcasts; NDP uses multicast (more efficient)
- Functions:
 - Find other devices on the network
 - Identify routers (router solicitation / router advertisement)
 - Duplicate Address Detection (DAD)

Router Solicitation (RS)

- Device sends a multicast asking any routers on the local subnet to respond
- Router replies with a Router Advertisement (RA) directly back to the device

Router Advertisement (RA)

- Sent in response to a router solicitation, or unsolicited to all devices on the network
- Unsolicited RAs are sent to a different multicast address than solicited ones
- RA provides local subnet information:
 - Prefix value
 - Prefix length
 - DNS server configuration
 - Other IPv6 configuration parameters

SLAAC (Stateless Address Autoconfiguration)

- Allows an IPv6 device to create its own routable IP address without a DHCP server

Process

1. Device sends Router Solicitation via NDP to determine the local subnet

2. Router responds with Router Advertisement containing the 64-bit subnet prefix
3. Device generates the last 64 bits (interface ID) using one of two methods:
 - Modified MAC address: inserts ff:fe in the middle of the MAC to create 64 bits
 - Random value: generates a random 64-bit interface ID
4. Device runs Duplicate Address Detection (DAD) via NDP to confirm no other device uses that address
5. Device now has a fully routable, unique IPv6 address

Duplicate Address Detection (DAD)

- Part of NDP
- Checks the network to ensure no other device is using the same IPv6 address
- Addresses the concern that self-assigned addresses could conflict with existing ones

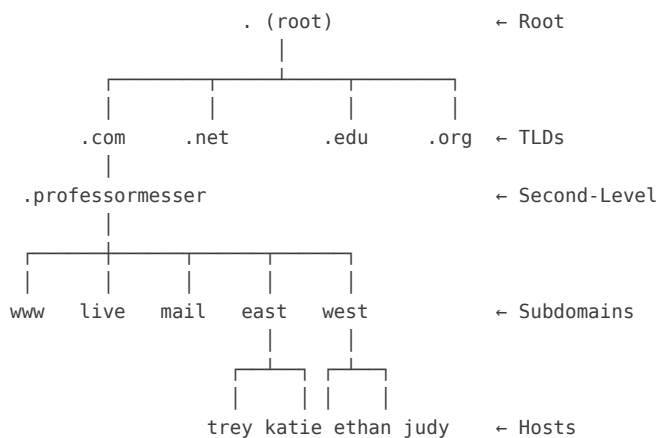
Purpose

- Translates domain names (e.g., `www.professormesser.com`) into IP addresses
- A hierarchical database of names

DNS Hierarchy

- Starts at the root (single dot `.`)
- Below the root: Top Level Domains (TLDs)
 - Generic: `.com`, `.org`, `.net`, etc.
 - Country codes: `.us`, `.ca`, `.uk`, etc.
- Below TLDs: domain names (e.g., `professormesser`)
- Full path from host to root = Fully Qualified Domain Name (FQDN)
 - Example: `www.professormesser.com`

DNS Hierarchy Diagram



DNS Infrastructure

- Root DNS servers: 13 server clusters, over 1,000 actual servers
- Root servers manage the TLDs
- Organizations can structure subdomains under their domain (e.g., `east.professormesser.com`, `west.professormesser.com`)

DNS Server Types

Primary DNS Server

- Contains all zone information for the DNS domain
- All configuration changes and updates happen here

Secondary DNS Server

- Receives zone information from the primary server (read-only)
- Provides redundancy — if primary is unavailable, secondary still resolves

- End users cannot tell which server answered their query

Local Name Resolution (Hosts File)

- Resolves names locally without querying a DNS server
- Use cases: testing servers, overriding incorrect DNS information
- File name: `hosts`
- Windows location: `windows/system32/drivers/etc/hosts`
- Simple text file mapping IP addresses to names
- Stored as read-only by default — must change permissions to edit
- Some applications ignore the hosts file and query DNS directly — check documentation

Forward and Reverse Lookups

Forward Lookup

- Provide a name → DNS returns an IP address
- Most common DNS operation

Reverse Lookup (Reverse DNS)

- Provide an IP address → DNS returns a name
- Must be separately configured on the DNS server
- The forward name and reverse name may differ

Authoritative vs. Nonauthoritative

Authoritative Server

- The primary DNS server for a DNS zone
- The authority for all records in that domain

Nonauthoritative Server

- A secondary server or caching server
- Returns cached information, not directly from the zone owner
- Cached information could be outdated if the authoritative server was recently changed

Time to Live (TTL)

- Configured on the authoritative DNS server
- Specifies how long (in seconds) a nonauthoritative server may cache a record
- After TTL expires, cached data is deleted and subsequent queries go to the authoritative server
- Example: TTL of 300 = information cached for a maximum of 5 minutes

Recursive DNS Queries

- Process by which a local DNS server resolves a name on behalf of the client
- The client (resolver) only contacts its local DNS server
- Local DNS server does all the work behind the scenes
- Results are cached for subsequent requests

Recursive Query Process

1. Resolver sends query to local DNS server (e.g., `www.professormesser.com`)
2. Local DNS server queries root server → root returns address of `.com` name server
3. Local DNS server queries `.com` name server → returns address of `professormesser.com` name server
4. Local DNS server queries `professormesser.com` name server → returns IP for `www.professormesser.com`
5. Local DNS server returns answer to resolver and caches it

DNS Security

Problems with Traditional DNS

- Traffic sent in the clear — anyone monitoring can see queried hostnames
- Responses are not authenticated — no way to verify responses are legitimate (could be spoofed)

DNSSEC (Domain Name Security Extensions)

- Digitally signs DNS responses
- Verifies responses came from a trusted source
- Confirms data has not been modified in transit
- Requires additional configuration on the DNS server
- Does NOT encrypt traffic — queries are still visible

DNS over TLS (DoT)

- Encrypts DNS traffic using TLS
- Uses TCP port 853
- Similar encryption to HTTPS web communication

DNS over HTTPS (DoH)

- Sends DNS queries over standard HTTPS
- Uses TCP port 443
- Traffic looks identical to normal encrypted web traffic in packet captures
- Some modern browsers use DoH by default

NTP (Network Time Protocol)

- Synchronizes clocks across all network devices (laptops, desktops, switches, routers, firewalls)
- Critical for comparing log files and timestamps from multiple devices
- Automatic process — devices update their clocks on a configurable schedule (daily, hourly, etc.)
- Accuracy: devices typically differ by only milliseconds

NTP Server

- Listens on UDP port 123
- Responds to time requests from NTP clients
- Does not modify its own time — uses a separate NTP client to query a different NTP server for updates
- Organizations can use external NTP servers or run their own internal NTP servers

NTP Client

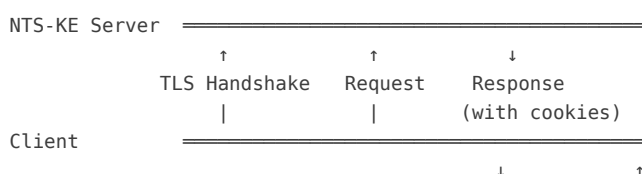
- Software that queries the NTP server to obtain time updates
- A single device can run both an NTP client and NTP server simultaneously

NTP Security Concerns

- By default, NTP sends information in the clear (unencrypted)
- Time is an important security concern:
 - Kerberos authentication fails if client and server timestamps differ by more than 5 minutes
 - An attacker providing wrong time could cause a denial of service

NTS (Network Time Security)

- Adds authentication to NTP so responses can be trusted
- Requires an NTS Key Exchange (NTS-KE) server on the network
- Two-step process:
 1. Client performs TLS handshake with NTS-KE server and receives a cookie
 2. Client includes the cookie in the NTP request to the NTP server to prove authentication
- NTP server responds with a trusted timestamp



PTP (Precision Time Protocol)

- Hardware-based time synchronization with nanosecond granularity
- Much more precise than NTP (which achieves 10 milliseconds on local servers)
- Used in industrial environments requiring very precise timestamps
- Requires separate dedicated hardware:
 - Has its own operating system
 - Runs without delays from third-party processes
 - Provides the most accurate timestamps possible

SSH (Secure Shell)

- Provides encrypted terminal/console access to remote devices (switches, routers, firewalls)
- All traffic encrypted — username, password, and all other data protected from capture
- Uses TCP port 22
- Replaces Telnet (TCP port 23) which provides same console-based view but no encryption
- Best practice: always use SSH, never use Telnet

Remote Desktop

RDP (Remote Desktop Protocol)

- Microsoft protocol for graphical remote control of Windows machines
- Access and use the desktop as if sitting in front of the monitor and keyboard
- Clients available for nearly any operating system

VNC (Virtual Network Computing)

- Uses RFB (Remote Frame Buffer) protocol
- Similar function to RDP but runs on many different operating systems
- Common choice for help desk/support teams needing to remotely control desktops

API (Application Programming Interface)

- Used to make changes to hundreds or thousands of devices
- Connects and controls devices using the language the device expects
- Allows automation with error handling capabilities
- More control than command-line scripts/batch files when problems occur

Console Connection

- Direct physical connection to the device via a separate console port
- Serial connection types: RJ45 serial, DB9 serial, or USB
- Text-based interface requiring knowledge of the device's CLI
- Works even when network connectivity is lost (cannot ping or SSH)
- Requires a laptop/desktop with a serial port or USB-to-serial adapter

Jump Server

- Single device you connect to first, then jump to other devices within the organization
- External connection via VPN tunnel or SSH
- Externally facing — must be highly hardened:
 - Multi-factor authentication required
 - Always kept up to date with security patches
 - High level of authentication to prevent brute force attacks
- Eliminates need to set up separate connections to each individual device

✦ SUPPLEMENTARY ✦

The “band” refers to the network itself. In-band = managing a device over the same network it carries traffic on. Out-of-band = a separate, dedicated connection (serial/USB/modem) independent of that network.

In-Band Management

- Connect to a device over the existing network using an IP address
- Device is assigned a management IP address, subnet mask, and networking details
- Access via SSH or web-based front end (internal web server)
- Management interface can be a separate interface or built into existing interfaces
- Example: switch with a dedicated 10/100/1000 management interface port

Out-of-Band Management

- Uses a serial interface that does not use the existing network
- Separate management/console interface via serial or USB connection
- Can connect a modem to the console/COM port for dial-in access via phone line
- Works even when the network is down
- COM server: dial into one server, then jump to other devices through that connection

Overview

- Encrypts data that would normally be sent in the clear across public networks (e.g., the internet)
- Allows secure communication over untrusted networks

VPN Concentrator

- Encrypts and decrypts VPN traffic
- Usually built into modern firewalls
- Can be dedicated hardware designed for encryption/decryption or software installed on an existing server

Client-to-Site VPN

- VPN client software installed on the user's workstation
- Client communicates back to a VPN concentrator at the corporate network
- Example: laptop at a coffee shop connecting securely to corporate
- Can be manually enabled/disabled or configured as always-on
- Flow:
 - Client encrypts → sends over internet → concentrator decrypts → forwards to internal network
 - Return traffic: corporate network → concentrator encrypts → sends over internet → client decrypts

Site-to-Site VPN

- Encrypts all communication between two sites over a VPN tunnel
- Commonly configured as always-on
- Built into existing firewalls at each location — VPN concentrator function enabled on both sides
- Users at remote sites are unaware encryption is occurring

Clientless VPN

- No separate VPN client software required
- Runs inside a browser using HTML5
- Uses the Web Cryptography API to provide encrypted tunnel within the browser
- Only requirement: an HTML5-compliant browser
- User visits the appropriate web page and the API provides VPN functionality

Full Tunnel vs. Split Tunnel

Full Tunnel

- All traffic from the client traverses the VPN tunnel
- No special forwarding decisions — everything goes through the VPN
- Third-party traffic (e.g., external websites) also goes through concentrator, gets decrypted, then redirected to the internet
- Return traffic from external sites goes back through the concentrator and VPN tunnel

Split Tunnel

- Some traffic goes through the VPN tunnel (corporate-bound)
- Other traffic (non-corporate) is sent directly to the internet without traversing the VPN
- Client workstation has two paths:
 1. VPN tunnel → concentrator → corporate network
 2. Direct internet → third-party websites
- VPN client recognizes destination and routes accordingly
- More efficient for third-party website communication while maintaining secure corporate access

AAA Framework

- **Identification** — public information (username, email address); does not grant access alone
- **Authentication** — proving identity with private information (password, authentication factor)
- **Authorization** — providing appropriate access to files, directories, network areas based on identity
- **Accounting** — tracking login/logout events, failed authentication attempts, other session details

Common AAA Authentication Flow

1. Client sends credentials (username, password, other factors) to a network device (e.g., VPN concentrator)
2. Device forwards credentials to AAA server
3. AAA server validates against stored credentials
4. If matched, server sends approval back to the device
5. Client gains network access

Single Sign-On (SSO)

- Authenticate once, access all authorized resources for the session
- Usually time-limited (often 24 hours)
- Not all authentication methods support SSO

RADIUS (Remote Authentication Dial-In User Service)

- Very common AAA protocol, around for a long time
- Supported across many operating systems and devices
- Used for VPN concentrators, server authentication, 802.1X wireless authentication
- Name references dial-in origins but works on modern networks
- Often a default option on firewalls, VPN concentrators, and other network devices

LDAP (Lightweight Directory Access Protocol)

- Protocol for reading and writing to a centralized network directory
- Provides context beyond simple username/password lists (departments, locations, organization)
- Based on ITU X.500 standard
- “Lightweight” version of the original Directory Access Protocol (DAP)
- Used by: Windows Active Directory, Apple OpenDirectory, Novell eDirectory

X.500 Distinguished Names

- Attributes provide hierarchical context for objects

Attribute	Field	Usage
CN	Common Name	Identifies the person or object
OU	Organizational Unit	A unit or department within the organization
O	Organization	The name of the organization
L	Locality	Usually a city or area
ST	State	A state, province, or county within a country
C	Country	The country's 2-character ISO code (e.g., c=US or c=GB)
DC	Domain Component	Components of the object's domain

- Example for a web server named WIDGETWEB:
 - CN=WIDGETWEB, OU=Marketing, O=Widget, L=London, DC=widget.com

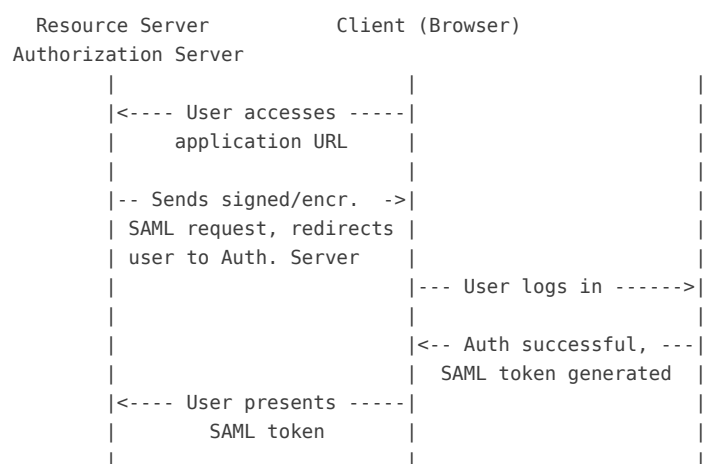
X.500 Directory Information Tree

- Hierarchical structure of objects
- **Containers** — higher-level grouping objects (countries, departments)
- **Leaf objects** — individual users or devices at the bottom of the tree

SAML (Security Assertion Markup Language)

- Open standard for authentication and authorization
- Not designed for mobile devices — limitation for multi-device scenarios
- Three components in SAML authentication flow:
 1. **Resource server** — holds the protected resource
 2. **Client** — user's browser
 3. **Authorization server** — validates credentials

SAML Authentication Flow



```
| - SAML token verified, ->|  
|   access granted   |  
|                   |  
|                   |
```

TACACS (Terminal Access Controller Access-Control System)

- Originally used to control access to dial-up lines on ARPANET
- Latest version: TACACS+
- Historically Cisco-centric but made public as an open standard in 1993
- Available for integration into any authentication system

Multifactor Authentication (MFA)

- Requires multiple authentication factors to prove identity
- Factor categories:
 - **Something you know** — password, PIN
 - **Something you have** — mobile phone, hardware token
 - **Something you are** — biometrics (fingerprint)
 - **Somewhere you are** — GPS location

TOTP (Time-based One-Time Password)

- Algorithm integrated into mobile authenticator apps
- Uses a pre-shared secret key + current time to generate pseudorandom codes
- Codes change periodically (typically every 30 seconds)
- Both client and server must be time-synchronized (via NTP)
- Used by Google, Facebook, Microsoft, and many others
- Falls under “something you have” factor

Compliance Overview

- Ongoing challenge: maintaining compliance with laws, policies, procedures, or rules
- Some compliance driven by type of business; others by state, local, or international laws
- Consequences of non-compliance: fines, incarceration, or loss of employment
- Scope: national, territory, state, domestic, or international requirements

Data Localization

- Data collected by a country must stay inside that country
- Must understand where data is supposed to be located and where it can move once stored

GDPR (General Data Protection Regulation)

- Regulation associated with the European Union
- Protects privacy of data associated with individuals residing in the EU
- Protected data includes: name, address, photo, email address, bank information, websites visited
- Requirements:
 - Data collected on EU citizens must be stored in the EU (data localization)
 - Users can decide where their data goes
 - Users can choose to have their data removed from sites
- Goal: give individuals control over their own data
- Often described as “right to be forgotten” — better described as a way for individuals to protect data they own

PCI DSS (Payment Card Industry Data Security Standard)

- Not a law — a standard created by the payment card industry
- Designed to protect credit card information
- Organizations are audited for compliance
- Non-compliance consequence: may lose ability to process credit cards

Six Areas of Focus

1. **Build and maintain secure networks and systems** — protect data moving across the network
2. **Protect cardholder data** — especially private information
3. **Maintain a vulnerability management program** — critical for organizations storing credit card data

4. **Implement strong access control measures** — limit who can access credit card data
5. **Regularly monitor and test networks** — ensure policies are working as expected
6. **Maintain an information security policy** — broader scope protecting all organizational data

States of Data

Data in Transit (Data in-Motion)

- Information transferred over a wired or wireless network
- Network infrastructure (switches, routers) designed to forward traffic, not protect it
- Security provided by other devices: firewalls, intrusion prevention systems
 - Monitor traffic, decide to forward or block, inspect for malicious content
- Encryption for data in transit: TLS (Transport Layer Security) and IPsec (Internet Protocol Security)

Data at Rest

- Information saved to a hard drive, SSD, or any other storage device
- Encryption types:
 - Full disk encryption (entire disk)
 - Database encryption (single part or entire database)
 - File/folder encryption (individual files or folders)
- Security policies applied via ACLs (Access Control Lists)
 - Determine which users have access or no access to data
 - System administrator or data owner controls permissions
 - Usually part of the operating system

PKI (Public Key Infrastructure)

- Policies and control of all encryption keys and certificates
- Not a small undertaking — significant work to create, with ongoing changes and updates
- Organizations often begin PKI when adding encryption keys to web servers or signing certificates
- Becomes a requirement for maintaining and managing all security assets organization-wide

Digital Certificates and Trust

Digital Certificates

- Assign a level of trust to a user or device
- Binds a public key with a digital signature and details about the key holder (name, organization, etc.)
- Digitally signed by a central Certificate Authority (CA) to add trust

Web of Trust

- Distributed form of trust (alternative to centralized CA)
- If A trusts B and B trusts C, then A can also trust C

Certificate Authority (CA)

- Centralized authority that all certificates start from
- Anything signed by the CA can be trusted anywhere in the organization
- CA digitally signs certificates — anyone can examine who signed and determine trust
- Options:
 - Internal CA: organization builds and self-signs its own certificates
 - Third-party CA: independent authority (useful for internet-facing trust)
- Certificate process built into Windows Domain services; third-party options available for other OS

IAM (Identity and Access Management)

- Process of managing permissions and access to data
- Components:
 - Access control: ensure users only access information needed for their job
 - Authentication: verify the user is who they claim to be
 - Authorization: grant appropriate permissions once authenticated
 - Auditing: track and monitor who accessed data and when

Least Privilege

- Users only have rights and permissions necessary for their job function — no additional access
- Reason administrator access is not assigned to every system
- Limits access to applications and data per job function

Role-Based Access Control (RBAC)

- Separate individuals into roles (e.g., shipping/receiving staff, manager, vice president)
- Administrator determines role definitions and associated permissions
- Users assigned to appropriate roles
- In Windows: implemented through groups — one group per role, users assigned to groups

Geographic Restrictions

Methods to Determine Location

- IP address (not always accurate)
- GPS (Global Positioning System)
- Wireless network name / access point location

Geofencing

- Allows or disallows access to information based on physical location
- Example: sensitive data only viewable from within corporate headquarters building
- VPN users may get different permissions based on connection location

Physical Security

CCTV (Closed Circuit Television)

- Cameras deployed around buildings/campus
- Modern features: motion detection, license plate reading, facial recognition
- All cameras networked to central storage for long-term retention
- Administrator can review any feed at any historical time frame

Door Locks

- Conventional lock: physical key, may include deadbolt
- Electronic reader: personal identification code (PIN)
- Token-based: RFID badge with electronic reader
- Biometric: hand print, fingerprint, retina scan
- Multi-factor: combining methods (e.g., badge + PIN)

Honeypots

- Virtual network with virtual servers/components designed to attract attackers
- Attacks are usually automated scripts/programs, not interactive humans
- Purpose: observe attacker behavior and techniques
- Many open-source versions available
- Constant battle: attackers are good at detecting what's real vs. fake

Honeynets

- Larger deception framework combining multiple honeypot components
- Includes: servers, workstations, routers, firewalls, switches, proxy servers, NAS
- Commonly virtual devices — can build any number to attract attackers
- Goal: make the environment seem real enough for attackers to move laterally

Risk

- Exposure to something harmful or dangerous
- Describes how possible it is for something bad to happen
- Constant concern for every organization; grows with organization size
- Must be considered when: expanding, adding applications, making configuration changes
- Used in business decisions: continue with a task or add security controls

Vulnerabilities

- Weaknesses in a system (e.g., operating system, application)
- If exploited, could allow unauthorized access to systems or data
- Undiscovered vulnerabilities may have existed for months or years
- Types:
 - Data injection
 - Authentication flaws
 - Exposed data
 - Security misconfiguration

Exploits

- When someone takes advantage of a vulnerability to gain access to a system or data
- Can be straightforward (e.g., embedded username in code) or complex (multi-step attack)
- Distinct from vulnerability: vulnerability = weakness, exploit = act of using it

Threats

- What is used by an attacker to exploit a vulnerability
- Can be intentional (attacker) or accidental (fire, flood)
- Internet threats often originate outside the organization

Threat Process

1. Operating system has a vulnerability
2. Threat agent creates a threat action
3. Threat action exploits the vulnerability
4. Result: system unavailability, data breach, or data exposure

CIA Triad

- Three fundamental principles of IT security
- Also called AIC triad (to differentiate from Central Intelligence Agency)

Confidentiality

- Protecting data from unauthorized access
- Methods: restricting access, encrypting data

Integrity

- Data has not been modified by unauthorized third parties
- Messages across network can't be modified without detection
- Commonly associated with digital signatures

Availability

- Systems and data remain accessible to authorized users
- Security measures must not impair overall system access

Network Segmentation Overview

- Physical separation of devices or logical/virtual segmentation (e.g., VLANs)
- Purposes:
 - **Performance** — isolate high-bandwidth applications onto their own network
 - **Security** — limit which devices can communicate with each other
 - **Compliance** — some regulations (e.g., PCI DSS) require segmentation for storing credit card data

IoT (Internet of Things) Segmentation

- IoT devices: sensors (cooling, heating, lighting), smart devices (home automation, video doorbells, cameras), wearables (smartwatch, health monitor)
- Problem: IoT devices may be ingenious in operation but poorly designed for security
- Solution: segment IoT devices onto their own network, away from production data

IIoT (Industrial Internet of Things)

- Machine-to-machine communication to keep industrial systems running
- Higher stakes than consumer IoT — entire manufacturing lines depend on proper communication
- Segmentation prevents external devices from disrupting critical communication
- Use cases: oil and gas systems, medical devices in hospitals
- Critical when lives are on the line

SCADA / ICS

- SCADA: Supervisory Control and Data Acquisition
- ICS: Industrial Control Systems
- Large industrial systems: manufacturing environments, power generation
- Require real-time monitoring and immediate network control
- Completely segmented from the rest of the network
- Only people with the right access in the right location can access these systems

Operational Technology (OT)

- Systems designed for critical infrastructure: electric grid, traffic control, manufacturing plants

- Uptime and availability are paramount
- Failure consequences:
 - Traffic lights malfunction or stop working
 - Manufacturing plants shut down completely
 - Power generation/distribution disrupted
- Network segmentation creates a stronger security posture for OT

Guest Networks

- Separate wireless network for visitors
- Allows internet access but prevents access to internal services
- Access controlled by passphrase or login authentication
- Segmentation keeps guest traffic isolated from internal network

BYOD (Bring Your Own Device)

- Personal mobile device used for work purposes (also called "bring your own technology")
- Device must meet organization's security requirements
- Segmentation separates personal data from company data on the same device
- Benefits:
 - Private information stays private
 - Company data managed by system administrator
 - If employee leaves: company data removed, personal data remains

Spoofing

- Pretending to be another person or device
- Examples: fake web server, fake DNS server, email address spoofing, caller ID spoofing
- Commonly used for on-path attacks — sitting in the middle of a conversation to monitor or change its contents

♦ SUPPLEMENTARY ♦

ARP = Address Resolution Protocol — resolves IP addresses to MAC addresses on a local network.

ARP Poisoning (IP Spoofing)

- Attacker pretends to be an IP address they are not
- Also called IP spoofing

How ARP Works

1. Device needs the MAC address of another device it wants to communicate with (only has the IP)
2. ARP sends a broadcast: "Who has [IP address]? I need that MAC address."
3. Device with that IP responds with its MAC address
4. Requester caches the response in a local ARP cache

The Vulnerability

- No authentication or security in the ARP process
- ARP accepts any response, even unsolicited ones

ARP Poisoning Attack

1. Attacker sends an ARP response to the victim — even without a request
2. Response spoofs the router's IP but contains the attacker's MAC address
3. Victim's ARP cache is updated with the attacker's MAC for the router's IP
4. All traffic from victim to router is now sent to the attacker first
5. Attacker forwards traffic to the legitimate router — neither victim nor router knows

Result

- On-path attack: attacker intercepts and forwards all traffic between victim and router
- Neither side realizes there's an attacker in the middle

DNS Poisoning (DNS Spoofing)

- Modifying information on the DNS server itself, or modifying DNS responses in transit
- Also called DNS spoofing

Methods

- **Modify the host file on the client** — host file has higher priority than DNS responses; entries always take precedence over DNS
- **Send fake responses to legitimate DNS requests** — requires changing information on the fly (on-path attack)
- **Hack into the DNS server** — change the DNS configuration directly

DNS Poisoning Attack Flow

1. DNS server contains the legitimate IP for a domain
2. Attacker either performs ARP poisoning to intercept DNS traffic, or hacks the DNS server directly
3. Attacker changes the stored IP to the attacker's IP
4. Subsequent DNS queries return the poisoned address
5. Victim's cache stores the spoofed IP
6. Victim communicates with the attacker's server instead of the legitimate one

DoS Overview

- Action or series of actions that causes a service to fail
- Often an overloading of a system so no resources are available to legitimate users
- Methods:
 - Overwhelm server capabilities
 - Exploit a vulnerability in an OS or application to cause failure
- Motivations:
 - Competitive advantage (competitors identified as attackers)
 - Distraction — keep troubleshooting resources busy while attacking another part of the network
- Physical DoS: pulling main power for an entire facility

Accidental Denial of Service

- Connecting switches without spanning tree protocol → network loop → traffic overwhelms switches
- Downloading a large file (e.g., Linux distribution) on limited-bandwidth internet connection
- Physical environment issues: water leak or roof leak taking out part of a data center

DDoS (Distributed Denial of Service)

- Multiple devices acting in unison to cause a denial of service
- Example: botnet takes over millions of personal computers, all directing traffic at one server
- Coordinated attack from devices located anywhere in the world
- **Asymmetric threat** — attacker uses very few resources to bring down systems with many more resources

DDoS Reflection and Amplification

- Attacker sends small amount of information; internet protocols multiply it into much larger traffic directed at victim
- Doesn't require many resources from the attacker
- Protocols used for amplification: NTP, DNS, ICMP

► EXAMPLE

DNS Amplification

- Normal DNS query: small request, small response (IP address)
- Amplified DNS query: request with ANY parameter returns much larger response (e.g., DNSSEC key information)
- 28-byte query → 1,300-byte response

DNS Amplification Attack Flow

1. Botnet command and control sends instructions to botnet
2. Botnet sends small DNS queries to open DNS resolvers on the internet
3. Botnet devices spoof the source address to be the victim's IP
4. DNS resolvers send amplified responses to the victim (spoofed address)
5. Amplified traffic overwhelms victim's web server resources

MAC Address Fundamentals

- MAC = Media Access Control — hardware address of an ethernet adapter
- Unique per device — allows traffic to be sent specifically to one device
- Format: 48 bits (6 bytes), displayed as hexadecimal with colon/period delimiters
- Structure:
 - First 3 bytes: OUI (Organizationally Unique Identifier) — identifies the manufacturer
 - Last 3 bytes: NIC-specific value — serial number of the adapter
- Stored in ROM (Read-Only Memory) of the adapter — called the burned-in address

MAC Address Table (Switch Learning)

- Switch builds a table of MAC addresses by examining source MAC addresses of inbound frames
- Table maps: source MAC address → switch interface
- Temporary cache — entries discarded after 5 minutes, must be relearned
- MAC addresses also used by STP (Spanning Tree Protocol) to avoid loops

Learning Process

1. Frame arrives at switch with a source MAC address
2. If source MAC is unknown, switch adds it to the table with the receiving interface
3. Future frames with that MAC as a destination are forwarded only to the mapped interface

Normal Forwarding

- Switch looks up destination MAC in the table
- If found → sends frame only out the mapped interface (directed conversation)
- Other devices on the switch do not receive the frame

MAC Flooding Attack

- Attacker sends many frames with different random source MAC addresses
- Fills up the limited MAC address table space
- When table is full and a destination MAC is not found, switch forwards frame to every interface
- Effectively turns the switch into a hub — all traffic sent to all ports
- Attacker can now capture all traffic on the switch, even traffic not destined for them

Why It Works

- Normal switch behavior: if destination MAC not in table, flood frame to all interfaces
- Guarantees traffic always reaches its destination
- Attacker exploits this normal process by filling the table with garbage entries

Mitigation

- Port security: limits how many MAC addresses can be learned on a single interface
- Makes filling the MAC address table much more difficult

Overview

- Broad term for malicious software that can cause problems, steal data, disrupt computers
- Capabilities: capturing keystrokes, capturing webcam images, forming botnets, displaying advertising, replicating to other devices

Malware Types

Virus

- Generic term for malware that uses humans to replicate and spread
- Requires human intervention to move from one machine to another (e.g., clicking a link)

Worm

- Malware that replicates from computer to computer without human intervention
- Considered much more dangerous than viruses — no click needed
- Relatively rare

Ransomware

- Encrypts all personal data on a system
- Attacker provides decryption key only after payment (usually cryptocurrency)
- At work: targets planning documents, employee PII, financial details
- Organizations and individuals often pay the ransom to recover data

Trojan Horse

- Malware that pretends to be something else (e.g., a game)
- Installs malware behind the scenes while appearing to function as expected

Rootkit

- Hides within the operating system itself
- Very difficult to detect and eradicate

Keylogger

- Records every key pressed — especially valuable during login (captures username/password)
- Sends captured credentials to the attacker

Adware / Spyware

- Presents advertisements on screen
- Advertisements make money for the malware author

Bloatware

- Not directly malicious — software preloaded on a system that is never used

- Takes up storage space and runs processes in memory
- Removing it frees resources for legitimate applications

Logic Bomb

- Waits for a particular event (time of day, day of week) before executing
- Dormant until trigger condition is met

Combined Malware Attack

► EXAMPLE

1. Worm exploits an unpatched vulnerability
2. Installs a remote access backdoor
3. Attacker installs a bot — can view and control the computer remotely

How Malware Executes

- Worms: run automatically without human action
- Clicking links in emails
- Pop-ups from websites enticing clicks
- Drive-by downloads: executables download without clicking anything on the page

Prevention

- Never click links embedded in emails
- Keep OS up to date with latest security patches
- Keep all applications updated
- Patching closes vulnerabilities that worms and other malware exploit

Rogue DHCP Server

- No security built into the DHCP protocol
- Any device can respond to DHCP requests — legitimate or not
- Damage: duplicate IP addresses, invalid IPs, loss of internet connectivity, network shutdown

Mitigation

- **DHCP snooping** — enterprise switch feature that examines all DHCP requests and only allows responses from legitimate servers
- **Active Directory authorization** — Microsoft AD determines which DHCP servers are authorized to hand out IPs
- If found: remove rogue server from network, then renew all IP addresses to ensure everyone has a legitimate IP

Rogue Access Points

- Access points are inexpensive and can be plugged into any ethernet connection
- May not be malicious — employee trying to expand wireless coverage
- Creates security issues: without proper security, anyone can access the network
- Can also be created via OS wireless sharing feature (turns a computer into an access point)

Detection

- Periodic network scans
- Physically walk the facility with a wireless analyzer

Mitigation

- **802.1X (Network Access Control)** — requires authentication before granting network access
- Even if a rogue AP is plugged in, users can't access the network without proper authentication

Wireless Evil Twin

- Malicious rogue access point designed to look exactly like legitimate APs at a location
- Combines phishing techniques with wireless networking technology
- Characteristics:
 - Same or very similar SSID (wireless network name)
 - Similar security settings
 - May duplicate captive portal configuration
 - Increased radio output power to overpower legitimate APs — becomes primary AP for the area

Mitigation

- Always use encrypted traffic (VPN or HTTPS)
- Even if connected to an evil twin, encrypted data can't be read

On-Path Attacks (Man in the Middle)

- Attacker sits in the middle of a conversation between two devices
- Receives information, examines it, possibly changes it, then forwards it
- Source and destination have no idea the attacker is in the middle or that data was changed

Types of On-Path Attacks

- Wireless evil twin
- ARP poisoning
- Session hijacking
- HTTPS spoofing
- Wi-Fi eavesdropping

General Mitigation

- Encrypt all data — even if attacker is in the middle, they cannot read the information

Phishing

- Social engineering with spoofing — pretending to be someone else to obtain personal information
- Indicators of phishing:
 - URL doesn't match the expected domain
 - Graphics slightly off
 - Spelling/grammar errors
 - Mismatched fonts
 - Generic greeting ("Dear User")
 - Missing punctuation
 - Sender's email domain doesn't match the claimed organization
- Goal: trick victim into entering credentials on a fake page, sending them directly to the attacker
- Never click links inside email messages

Shoulder Surfing

- Viewing someone's screen in a public location (airport, restaurant, coffee shop)
- Can also be done from a distance using binoculars or a telescope from another building
- Advanced version: malware enabling the victim's camera to watch them remotely

Mitigation

- Be aware of surroundings — sit with back to wall
- Avoid viewing sensitive information in public
- Use privacy filters on LCD screens — display appears black from side angles
- Position monitors away from windows (also helps with glare)

Tailgating

- Following an authorized person through a secured door without their knowledge
- Example: catching a door before it closes after someone badges in and walks away

Piggybacking

- Authorized person knowingly lets an unauthorized person in
- Example: carrying donuts/lunch and asking someone to hold the door

Tailgating/Piggybacking Mitigation

- Visitor badge policy — visitors must wear visible badges

- Challenge people without badges — ask where their badge is
- One-person-at-a-time badging — physically close door between each person
- Access control vestibule (airlock) — mechanical enforcement allowing only one person at a time
- Train employees to look for things out of the ordinary

Dumpster Diving

- Going through an organization's trash to find sensitive information
- Garbage bins often open and unlocked
- Information found: names (for impersonation/phishing), contact information, project details
- Best timing: end of a quarter or end of a big project
- Legality (US): generally legal if trash has been thrown out, unless on private property with no-trespassing signs or local/state regulations prohibit it

Mitigation

- Lock up garbage bins
- Fence around the area
- Monitoring cameras
- Shred sensitive information (or use third-party shredding service)
- Burn sensitive documents (common in government)

Overview

- VLANs separate different parts of a network (e.g., marketing, accounting, shipping/receiving; or home: normal traffic, IoT, cameras)
- Devices on one VLAN cannot communicate with another VLAN without a router
- VLAN hopping bypasses this restriction without a router
- Two methods: switch spoofing and double tagging

Switch Spoofing

- Exploits switch autoconfiguration (trunk negotiation)
- Switch automatically determines if connected device is another switch
- If detected as a switch, it configures a trunk connection between interfaces
- Attacker pretends to be a switch → gains trunk access → can send traffic to any VLAN on that port

✦ SUPPLEMENTARY ✦

A **trunk** is a switch port configured to carry traffic for multiple VLANs simultaneously between switches using 802.1Q tags. Contrast with an **access port**, which carries traffic for a single VLAN and connects to end devices.

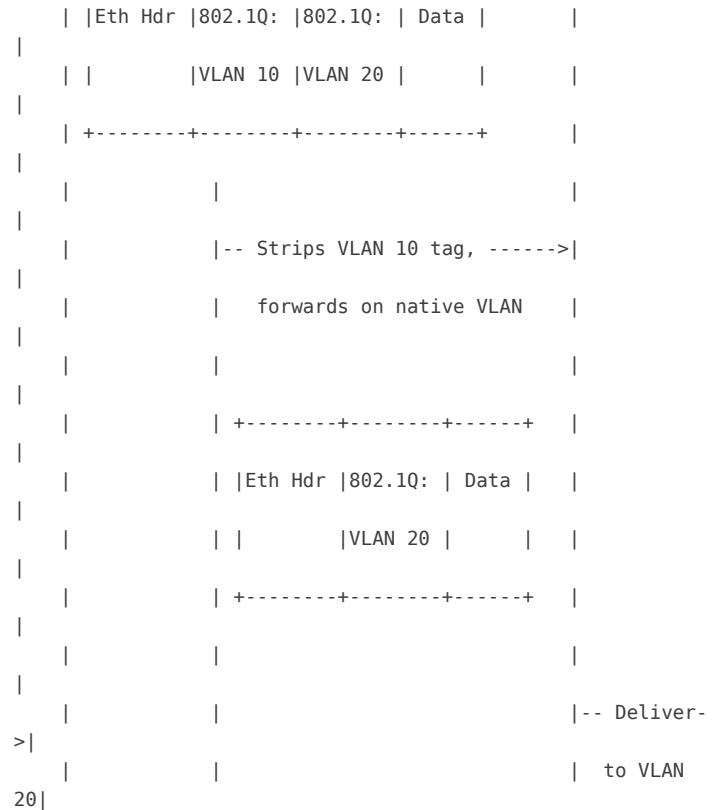
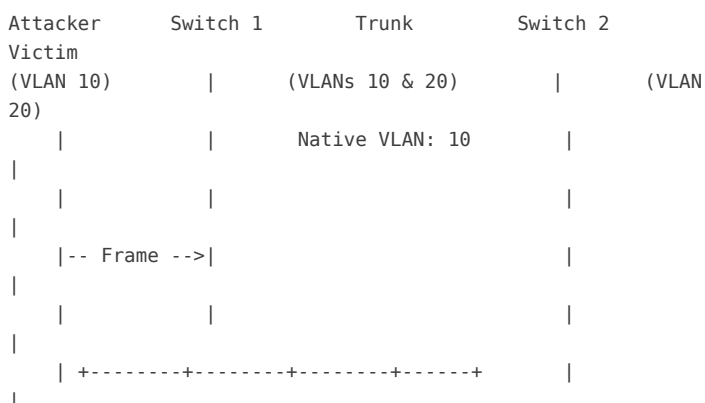
Mitigation

- Disable trunk autonegotiation
- Manually configure trunk interfaces
- Configure which specific VLANs are allowed to pass between switches

Double Tagging

- Exploits the native VLAN configuration on switches
- Attacker crafts a frame with two 802.1Q VLAN tags
- One-way attack only — no response can be received (useful for DoS)

Double Tagging Diagram



How Double Tagging Works

1. Attacker (on VLAN 10) creates a frame with two tags:
outer tag = VLAN 10, inner tag = VLAN 20
2. First switch reads the outer VLAN 10 tag, removes it,
forwards frame onto VLAN 10 (native VLAN on the trunk)
3. Frame arrives at second switch with only the VLAN 20 tag
remaining
4. Second switch interprets it as a trunked frame for VLAN
20, removes the tag, delivers to victim on VLAN 20

Requirements for Double Tagging

- Attacker must be on the same VLAN as the trunk's native VLAN
- Native VLAN is typically VLAN 1 by default

Mitigation

- Don't place users on the native VLAN
- Change the native VLAN ID from the default (VLAN 1) to another value
- Force tagging of the native VLAN for all trunk traffic

Disabling Unused Ports (Network Services)

- Every network-based service has an open port number visible to others on the network
- Open ports are entry points into services running on the system
- If a service is no longer in use: close it and ensure the port is not externally accessible
- Port range: 0–65,535
- Use a firewall to control which devices can connect to open ports
- Use tools like Nmap to scan for open ports and decide which to keep or close

Changing Default Credentials

- Switches, routers, firewalls, and other devices ship with standard default credentials
- Many devices don't force you to change them
- Default credentials often provide full administrative access
- Leaving defaults allows anyone who knows them complete control of the system
- Database of default credentials available at routerpasswords.com

Port Security (Switch Feature)

- Prevents unauthorized devices from connecting to a switch interface
- Based on MAC address of connected devices
- Configuration options:
 - Set how many MAC addresses are allowed per interface
 - Specify exact MAC addresses allowed on an interface
 - Switch monitors and records MAC addresses of connected devices
- If unexpected device connects: port security activates
- Default action on many switches: immediately disable the interface and notify the network administrator

Disabling Unused Switch Interfaces

- Best practice: disable interfaces until someone needs them
- Requires additional administration (tracking enabled/disabled state)
- Makes the network much more secure

Network Access Control (NAC) / 802.1X

- Requires authentication before any device can communicate on the network
- Applies to wired and wireless connections
- User must provide username and password before gaining network access

MAC Address Filtering

- Limits network access based on hardware (MAC) address of devices
- Useful for keeping unauthorized devices off the network
- Challenge: MAC addresses can be administratively changed (software configuration)
- Attacker can capture packets, find valid MACs, wait for one to leave, then spoof it
- Categorized as **security through obscurity** — knowing the method makes it easy to circumvent

Key Management

- Third-party software to manage authentication details, certificates, encryption keys, SSH keys
- Capabilities from a single console:
 - Create keys for specific services or cloud providers
 - Associate keys with specific users or services
 - Track expiration dates
 - Renew or revoke keys
 - Monitor key usage and access
- Manages: SSL certificates, SSH keys, license details, certificate authorities
- Generates reports: SSH reports, landing server reports, private key reports

Access Control Lists (ACLs)

- List of traffic that is allowed and traffic that is disallowed
- Criteria can include: source IP, destination IP, port number, time of day, application
- Can combine criteria into complex rules
- Can group IP addresses — some allowed, some denied
- Found on: routers, firewalls, operating systems, anything that makes access decisions

Firewall Rules

- Complex form of an ACL
- Components: rule name, source/destination zone, source/destination address, destination port, username
- Very specific and fine-grained security controls

Rule Processing

- Interpreted top-to-bottom — first match wins
- More specific rules placed at the top
- More general rules placed lower
- **Implicit deny** — if no rule matches, traffic is automatically denied (no explicit rule needed)

► EXAMPLE

Rule Set

Rule	Remote IP	Remote Port	Local Port	Protocol	Action
1	All	Any	22 (SSH)	TCP	Allow
2	All	Any	80 (HTTP)	TCP	Allow
3	All	Any	443 (HTTPS)	TCP	Allow
4	All	Any	3389 (RDP)	TCP	Allow
5	All	53 (DNS)	Any	UDP	Allow
6	All	123 (NTP)	Any	UDP	Allow
7	All	Any	Any	ICMP	Deny

♦ SUPPLEMENTARY ♦

Rules 1–4 are **incoming** (allowing outside devices to connect to local services). Rules 5–6 are **outgoing** (allowing this server to make DNS/NTP requests). Rule 7 is **incoming** (denying inbound ICMP/ping).

Content Filtering

URL Filtering

- Filter traffic based on specific URLs or categories of URLs (auction, hacking, travel, recreation, etc.)
- Also called URI (Uniform Resource Identifier) filtering
- Uses Allow lists and Block lists

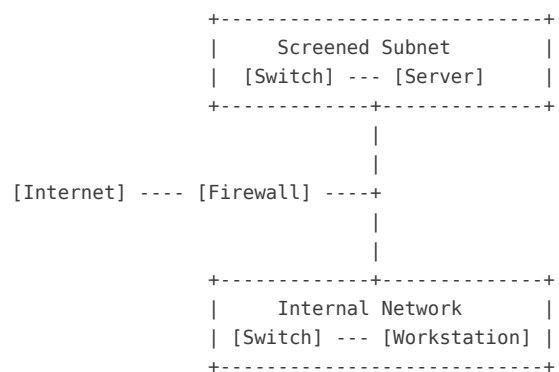
- Categories are easier to manage than individual URLs
- Users try to circumvent — combine with firewall rules
- Most next-generation firewalls have URL filtering built in

Other Content Filtering

- Filter based on data content: internal documents, financial details
- Prevent non-safe-for-work content
- Parental controls at home
- Antivirus/antimalware — filters malicious software in network traffic

Screened Subnet

- Separate area of the network for services made available to the public
- Hosts public web servers, public email servers
- Visitors directed to screened subnet, away from internal services
- Internal network remains isolated — internal users can still reach the internet



Security Zones

- Broad labels used in firewall rules instead of IP address ranges
- Simplify security policies — rules reference zone names instead of specific addresses

Examples of Zones

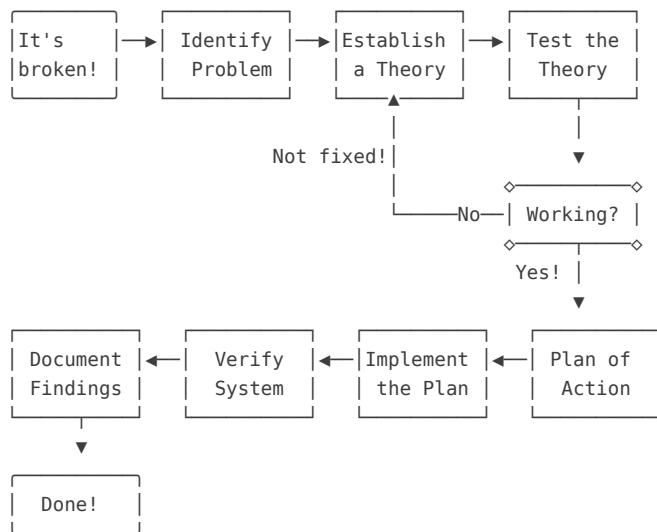
- Trusted / Untrusted
- Internal / External
- Inside / Internet / Server / Databases / Screened subnet

Zone-Based Firewall Rules

- “Trusted zone can communicate to untrusted zone” — no IPs or ports needed
- “Untrusted zone can access screened subnet”
- “Untrusted zone denied from trusted zone”
- More zones = more granularity in firewall rules

Overview

1. Identify the problem
2. Establish a theory of probable cause
3. Test the theory
4. Establish a plan of action
5. Implement the solution
6. Verify full system functionality
7. Document findings



Step 1: Identify the Problem

- May be obvious (e.g., severed cable) or require investigation
- Attempt to duplicate the issue
- Talk to users — gather their experiences and observations
- Combine user reports with statistics/metrics from routers and switches
- Consider multiple symptoms working together
- Ask: what has changed since it last worked?
 - Cables moved in wiring closet
 - System powered off
 - Configuration changes
- Build a lab to duplicate the problem — easier to find root cause
- Break large problems into smaller components

Step 2: Establish a Theory of Probable Cause

- Start with obvious/quick-to-solve problems (e.g., swap a cable)
- Use OSI model approaches:

- **Top-down** — start at application layer, work down (good for existing networks)
- **Bottom-up** — start at physical layer, work up (good for new implementations)
- Eliminate variables — if issue appears on multiple OSes, OS is not the cause

Step 3: Test the Theory

- Execute steps to confirm/deny the theory
- Make changes in a lab environment and evaluate the effect
- If theory is confirmed — proceed to plan of action
- If theory is not confirmed — return to Step 2 and establish a new theory

Step 4: Establish a Plan of Action

- Determine how to implement the fix in production
- Some changes can be made during business hours; many require change control
- Prepare contingencies:
 - Plan B (and Plan C) if primary plan fails
 - Rollback process to revert to previous state

Step 5: Implement the Solution

- Apply the fix during the assigned change control window
- Implementation team may differ from troubleshooting team
 - Troubleshooting team determines the fix
 - Operations team implements the change

Step 6: Verify Full System Functionality

- Confirm with end users that the problem is resolved
- Not confirmed until users verify
- Discuss preventive measures with users
 - Users may suggest workflow improvements
 - IT may suggest technology-based prevention

Step 7: Document Findings

- Document the process followed
- Document the specific change that resolved the issue
- Store in help desk database or knowledge base
- Enables future reference if the problem recurs

Fiber Optic Mismatches

- Multimode fiber — light uses multiple paths (modes) through the fiber
- Single mode fiber — light uses a single path through the fiber
- Fiber core sizes:
 - Multimode: 50 micron or 62.5 micron (cladding brings total to 125 micron)
 - Single mode: 9 micron (cladding brings total to 125 micron)
- All three types feel identical in hand (125 micron total diameter)
- Mixing fiber types causes signal errors and communication failure
- Fiber often color-coded, but cannot always rely on color
- Check writing on the outside of the fiber for type/size documentation

Copper Cable Categories

- Cable construction standardized by TIA (Telecommunications Industry Association)
- TIA sets manufacturing standards and tests for qualification
- Cables qualify as a specific category after testing
- Different categories support different signal types/speeds
- IEEE ethernet standards specify minimum cable category:

Ethernet Standard	Cable Category	Maximum Supported Distance
1000BASE-T	Category 5	100 meters
1000BASE-T	Category 5e (enhanced)	100 meters
10GBASE-T	Category 6	Unshielded: 55m / Shielded: 100m
10GBASE-T	Category 6A (augmented)	100 meters
10GBASE-T	Category 7 (shielded only)	100 meters
40GBASE-T	Category 8 (shielded only)	30 meters

Bandwidth vs. Throughput

- **Bandwidth**
 - theoretical maximum data rate over a link (bits per second)
 - the size of the pipe
- **Throughput**

- actual amount of data transferred in a time frame (bits/sec or bytes/sec)
- how much water is flowing through the pipe
- May need to convert between bits and bytes when comparing

Distance Limitations

- Every cable standard has a maximum distance while maintaining signal quality
- Part of the IEEE standard specification
- Example: 1000BASE-T = Cat 5, 100 meters max

Cable Testing

- Use a cable tester to validate installed cables
- Tests similar to TIA category qualification tests
- Tester reports which category the installed cable matches
- Wrong category = errors on the link = slowdown
- Physical errors often caused by signal attenuation or complete signal loss
- CRC errors indicate physical layer problems

UTP vs. STP

- **UTP (Unshielded Twisted Pair)**
 - four pairs of wires, no shield, each pair twisted
- **STP (Shielded Twisted Pair)**
 - shield around entire cable and/or individual pairs
 - Includes grounding wire for the shield
 - More resistant to electrical interference
 - Must peel back shielding to terminate into RJ45 or punchdown block

Crosstalk (XT)

- Signal from one pair leaking into another pair of wires
- Types of crosstalk measurements:
 - **NEXT (Near-End Crosstalk)** — measured at the end closest to the testing device; usually strongest signal point
 - **FEXT (Far-End Crosstalk)** — measured at the far end of the cable
 - **Alien crosstalk** — interference from other nearby cables
 - **ACR (Attenuation to Crosstalk Ratio)** — compares signal (insertion) loss vs. near-end crosstalk; effectively a signal-to-noise ratio
- Signal-to-noise ratio:

- 10:1 = signal is 10x stronger than noise (good)
- 1:1 = signal equals noise (bad)
- Reducing crosstalk:
 - Reconnect/re-terminate the cable
 - Maintain as many twists as possible (untwist minimally for termination)
 - Cat 6A uses larger cable with spacer between pairs
 - Test cables to verify category compliance
- Cable tester easily identifies crossed pairs
- **Auto-MDIX** — ethernet chipset detects cable cross and reverses it electronically
 - Not always enabled or supported on every chipset
 - Best practice: properly terminate cables rather than relying on Auto-MDIX

EMI (Electromagnetic Interference)

- Affects both UTP and STP cables
- Maintain shield integrity on STP — any break allows EMI in
- Respect minimum bend radius (check cable documentation)
- Never use staples on network cables
- Limit plastic cable ties; use Velcro instead to avoid crimping wires
- Avoid running near:
 - Power cords
 - Fluorescent lights
 - Electrical systems/generators
 - Fire prevention components
- Test cable signal-to-noise ratio if EMI is suspected

Attenuation

- Loss of signal strength over distance
- Occurs in both copper and fiber optic connections
- Same concept as wireless signal weakening with distance from antenna
- Reason for distance limitations in IEEE standards

Cable Termination Issues

- Connectors look similar but may have different specifications
- Quality varies between installers
- Use someone well-versed in network cable installation standards
- Pin-to-pin matching: pin 1 to pin 1, pin 2 to pin 2, etc.
- Mismatched pins can cause:
 - Speed drops (e.g., expecting 1 Gbps but only getting 100 Mbps)
 - Complete signal loss (e.g., pin 1 connected to pin 2 and vice versa)
- **Crossed pairs** — pin 1 connects to pin 3, pin 3 to pin 1 (half the wires crossed, half straight)
 - Can occur at RJ45 crimp or punchdown block
 - Check every connection along the path

Power over Ethernet (PoE)

- Provides power over the same wire used for data
- Simplifies installation — no separate power cable needed
- Common devices: telephones, access points, cameras

PoE Power Sources

- **Endspan** — power built into the switch itself (PoE switch)
- **Midspan** — separate device (injector) between the switch and PoE device that injects power onto the wires

PoE Standards

Standard	DC Power Output	Maximum Current	Typical Devices
PoE	15.4 watts	350 mA	Simple telephones, small access points
PoE+	25.5 watts	600 mA	Larger telephones, cameras
PoE++ Type 3	51 watts	600 mA	Laptops, PTZ cameras
PoE++ Type 4	71.3 watts	960 mA	Laptops, PTZ cameras

- PoE++ supports additional Ethernet standards: 2.5-gigabit, 5-gigabit, and 10-gigabit

PoE Compatibility

- Switch and device must use compatible PoE standards
- A PoE+ switch cannot power a PoE++ device
- Some switches have all interfaces PoE-capable; others have a mix of PoE, PoE+, and PoE++ ports
- Switches have a maximum total PoE power budget (e.g., 200W or 720W)
- Must add up all connected devices' maximum power draw and ensure it is under the switch's capacity

Transceiver Mismatches

- Transceivers provide modular connections for Ethernet devices
- Fiber transceivers must match the type of fiber being connected
- Wavelength marked on the transceiver (e.g., 850 nm or 1310 nm)
- All wavelengths must match throughout the entire link
- Mismatched transceivers cause signal loss, increasing error counters, loss of signal, or network slowdown
- Transceivers look nearly identical — wavelength markings are very small

- Once plugged into a switch, markings are no longer visible — may need to check switch specifications or physically remove the transceiver to read the label

Transceiver Signal Strength (Power Budget)

- Each device has a **sensitivity level** — minimum signal it can receive and still interpret properly
- Important for long runs with many connections in the middle

Power Budget Calculation

1. Determine transmitted power (measured in dBm — decibels per milliwatt)
2. Calculate signal loss from:
 - Media distance (fiber length)
 - Connectors and splices (each reduces signal)
3. Subtract total signal loss from transmitted power = received power
4. Compare received power to receiver sensitivity value

Interpreting Results

- Sensitivity values are negative (signal decreases through media)
- If received power $\geq$ sensitivity value → good signal
- If received power $<$ sensitivity value → insufficient signal
- Example: transceiver with receiver sensitivity of -17 dBm
 - Received power of -17 dBm or higher (e.g., -14 dBm) → good
 - Received power of -20 dBm → not enough signal

Interface Monitoring

- Network administrators monitor interfaces on important devices to detect developing problems
- Allows resolution before an outage occurs
- Can warn of congestion or overutilization — may drive network design changes
- Operating systems provide interface performance feedback under network configurations
- Administrators automate monitoring using SNMP (Simple Network Management Protocol)
- Many devices support MIB-II (Management Information Base) — a standard set of SNMP statistics common across devices
- Proprietary MIBs available for device-specific statistics (firewalls, switches)

Key Interface Metrics

- **Link status**
 - Whether the link is up or down
 - May indicate cable problem, interface issue, or device reboot
- **Utilization**
 - Total throughput over the connection
 - Ensure sufficient bandwidth for all services
 - Run bandwidth tests to determine actual throughput capacity
- **Errors**
 - CRC errors, runts, giants, drops
 - Often points to cable or interface problems

Ethernet Frame Structure

Field	Purpose
Preamble + SFD	Identifies beginning of frame (not visible in packet captures)
Destination MAC	Where the frame should go (used by switches for forwarding)
Source MAC	Device sending the frame
EtherType	Describes what type of data is in the payload
Payload	The actual data being transmitted
Frame Check Sequence	CRC checksum to verify data integrity

CRC Errors

- Frame Check Sequence contains a checksum calculated at transmission

- Receiving adapter recalculates CRC and compares to the FCS value
- If they match — frame received without corruption
- If mismatch — CRC error counter incremented
- First warning of signal problems on a connection
- Indicates possible cable or interface issue

Runts

- Frame received that is less than 64 bytes (minimum ethernet frame size)
- Frames below 64 bytes are counted as errors
- Rare on modern full-duplex switch networks
- Common on half-duplex networks during collisions

Giants

- Frame received larger than 1,518 bytes (default maximum ethernet frame size)
- Jumbo frames can exceed 1,518 bytes but have their own configured maximum
- If a frame exceeds the configured maximum size, it is classified as a giant

Drops

- Frames lost due to contention or buffering problems
- No room in buffer to hold the frame
- Drop counter increments on the system
- Indicates communication problems on the network

Viewing Error Counters

- Most devices/operating systems provide a way to view these counters
- Cisco example: `show interface` command displays runts, giants, CRCs, and other errors
- Slowly incrementing CRC errors may indicate developing cable/interface problem

Error Disabled State

- Device automatically disables an interface without human intervention
- Common triggers:
 - **Interface flapping** — link going up/down repeatedly, causing spanning tree issues
 - **Port security violation** — unauthorized device connected to a secured port
 - **Configuration problems** — incompatible settings

- **Increasing error counts** — too many errors detected
- Once error disabled, the interface stays off until manually re-enabled
- Must log into switch and administratively re-enable the interface
- If same problem recurs, interface may return to error disabled state

Administratively Down

- An administrator intentionally disabled the interface
- Not automatic — deliberate action by a human
- Must log in and administratively enable to restore

Suspended Port Status

- Error occurs the moment the interface is enabled
- Interface connected to an incompatible configuration
- Example: LACP configured on one switch but not the other — interface immediately suspends

Routing Tables

- Used to determine the best next hop when forwarding traffic through a router
- Builds a map of where data will be forwarded
- Contains: default gateway configurations, static routes, dynamically learned routes
- If no route exists for a destination, the router drops the traffic
- Router may send back an ICMP host unreachable message to the sender
- When troubleshooting, check the routing table of every router along the path
- Must confirm routes in both directions — to the destination and back to the source
- Helpful to compare routing tables against a network map

Gateway of Last Resort

- A default route that summarizes all possible destinations into one entry
- Used when no other route in the routing table matches the destination
- Usually added as a static route by the administrator
- Destination is 0.0.0.0/0 — encompasses every host on every network
- If nothing else matches in the routing table, the gateway of last resort is used

► EXAMPLE

Routing Table

- Directly connected routes (e.g., 10.10.10.0/24, 10.10.40.0/24, 10.10.50.0/24)
- Static routes (e.g., 10.10.20.0/24)
- Dynamically learned routes via RIP (e.g., 10.10.30.0/24)
- Gateway of last resort configured as a static route to 0.0.0.0/0 via a next-hop address

Address Pool Exhaustion

- If DHCP address pool is exhausted, devices receive an APIPA address instead
- APIPA allows communication on the local subnet only — nonroutable
- Cannot communicate outside the local subnet with APIPA

♦ SUPPLEMENTARY ♦

APIPA = Automatic Private IP Addressing (range 169.254.0.0/16)

Resolution

- Check DHCP server for available IP addresses
- Add additional addresses to the pool if needed
- Use IPAM (IP Address Management) to monitor pools and availability across DHCP servers
- Decrease lease time if users connect for short periods — frees up addresses faster and minimizes exhaustion

Incorrect IP Configuration

- Devices may receive wrong IP address, subnet mask, or default gateway
- Confirm received values are correct for the connected interface
- Perform a packet capture to see what other devices on the subnet are configured as
- Check configuration of other devices already on the same network
- Troubleshooting order: ping local address → ping default gateway → ping an address beyond the gateway
- Use ping and traceroute to map position in the network and verify correct assignment

Duplicate IP Addresses

Common Causes

- Manually configured device using an address that exists in a DHCP pool
- Multiple DHCP servers with overlapping pools handing out the same addresses
- Rogue device added to network with DHCP enabled, handing out unexpected addresses

Detection

- Older operating systems: two devices fight over priority based on switch MAC address table
- Modern operating systems: duplicate IP discovered when device first connects; prevented with an error message displayed on screen

Troubleshooting

- For manually configured devices: check assigned IP address, subnet mask, and default gateway
- Ping the IP address before statically configuring it — a response means it is already in use
- If unexpected device responds to ping: check ARP table to get its MAC address
- Use the switch MAC address table to find which interface the device is plugged into

- For overlapping DHCP pools: perform a packet capture to see what DHCP servers are offering

Switching Loops

- At the MAC address level, there is no mechanism for counting how many times a frame is seen (no TTL equivalent)
- Frames in a loop will circle indefinitely until a connection is unplugged
- Broadcasts and multicasts amplify the problem — one frame transmitted out every other interface
- IP has a TTL field to prevent packets from looping, but there is nothing in the frame at Layer 2 to prevent loops
- Adding a second connection between switches accidentally creates a loop
- Looping traffic quickly overwhelms both switches and brings the network to a halt

Spanning Tree Protocol (STP)

- Prevents loops on the network
- Uses Bridge Protocol Data Units (BPDUs) — MAC layer multicast frames sent between switches
- BPDUs contain configuration details and inform of topology changes
- All switches on the local broadcast domain see these multicasts

BPDUs Timers

- BPDUs sent every 2 seconds by default (hello timer)
- If three consecutive hellos are missed (6 seconds), the link is considered down
- Spanning tree then redesigns the topology to prevent loops

Root Bridge Election

- Root bridge elected when the network first starts
- All bridges participate in the election
- Each switch chooses the best connection to communicate to the root bridge
- Root bridge has the lowest bridge ID
- Bridge ID can be manually set between 0 and 61,440
- If multiple switches have the same bridge ID, the lowest MAC address becomes root bridge

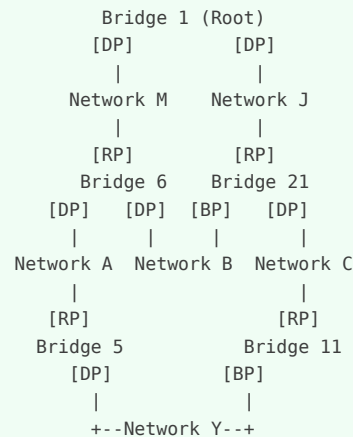
STP Port Roles

- **Root port** — the port on a non-root switch used to reach the root bridge
- **Designated port** — active port passing traffic
- **Blocked port** — blocked by spanning tree to prevent loops

EXAMPLE

STP Topology

Legend: [RP] Root Port [DP] Designated Port [BP] Blocked Port



- Bridge 1 is root (lowest bridge ID)
- Blocked ports on Bridge 21 (toward Network B) and Bridge 11 (toward Network Y) prevent loops
- If Bridge 6's link to Network M fails, spanning tree unblocks alternate paths to restore connectivity

STP Port States

State	Function
Blocking	Not passing traffic; preventing loops
Listening	Listening for other switches on the broadcast domain before changes
Learning	Adding information to the MAC address table
Forwarding	Actively passing traffic
Disabled	Administratively turned off by the network administrator

STP Topology Change

- When a link fails (three missed hellos), spanning tree reconfigures the network
- Blocked ports are unblocked to provide alternate paths
- Network continues to function without loops through the new topology

VLAN Assignment Issues

- Every switch interface is associated with a particular VLAN
- Access ports are assigned a single VLAN ID
- A device with an IP address that cannot communicate to others on the same network may have a VLAN misconfiguration
- Must check switch configuration to determine which VLAN is assigned to a physical interface

- Resolution: change the VLAN ID on the interface, or move to an interface already configured for the correct VLAN
- Very common issue on networks with many VLANs; resolved relatively quickly

Access Control Lists (ACLs)

- Even with correct VLAN and routing configuration, ACLs may block traffic
- Check switches and routers for ACLs configured on interfaces
- Should be part of normal troubleshooting when no traffic can communicate between networks

ACL Best Practices

- More granular rules at the top of the list (ACL stops evaluating after a match)
- More common matches at the top for efficiency
- Broader controls lower in the ACL
- Disable ACL functionality before making changes (possible to lock yourself out)
- Default behavior on many devices: adding an ACL to an interface implicitly denies all traffic not specifically listed
- An empty ACL effectively filters all communication through that device

Congestion

- Networks run at a predefined speed (e.g., 1000BASE-T operates at 1 Gbps)
- Congestion occurs when multiple links send more traffic than the destination link can handle
- Example: two 1-Gbps links sending to a single 1-Gbps destination — cannot fit 2 Gbps into 1 Gbps
- Excess packets queue into a buffer, but buffers in switches/routers are relatively small
- Once the buffer fills, packets are discarded
- Resolution: increase the size/speed of the network, or decrease the amount of traffic

Bottlenecks

- “The network is slow” usually means a bottleneck somewhere is causing a slowdown
- Can be difficult to troubleshoot — problem may involve many different technologies:
 - System bus speed
 - CPU speed in a switch or router
 - Storage drive speed (HDD vs SSD)
 - Network link speeds across different locations
- Must examine all parameters between devices to find the bottleneck
- Sometimes obvious, often requires drilling into system details
- The slowest link between two devices is likely the bottleneck holding up throughput for all other networks

► EXAMPLE

Bottleneck

- Web transaction response times around 1,500–1,750 ms (almost 2 seconds of delay)
- Database was consuming most of the time
- Configuration changes to the database server eliminated the bottleneck
- Response times dropped to around 500 ms

Bandwidth and Throughput

- **Bandwidth percentage** — measure of how much a network is being used over a particular amount of time (presented as a percentage)
- **Throughput** — how much data was moved through the network during a time frame

- Methods to monitor bandwidth: gather directly from a switch, use SNMP, or use NetFlow

Latency

- The delay between the request and the response
- There will always be some latency — it takes time to move information between devices
- Ideally measure response times at every stop along the path
- Allows breaking down delays into smallest parts per segment
- Requires measuring tools in every network link along the path
- Packet captures provide microsecond granularity — shows exactly how long a packet stayed in a device, traversed a network, and forwarded to the next segment

Packet Loss

- Ideal: all traffic makes it across the network all of the time
- Packet loss/discard: no errors with the packet, but discarded for other reasons
- Causes:
 - Network outage
 - Contention — not enough bandwidth
 - Bad wireless network
 - Bad cable causing corruption — corrupted data is identified at the other side, discarded, and must be re-sent
- Re-sending takes additional time and resources, causing significant application delays

Jitter

- VoIP calls and live video streams are very sensitive to delay
- Packets should arrive at regular, predictable intervals
- If a packet is lost during real-time communication, it cannot be retransmitted — must be discarded and continue forward
- Effects: clicking noise on phone calls, stutter on live video streams
- **Jitter** — the variation in time between received frames
- Low jitter: packets received at regular intervals with small, consistent delay between them
- High jitter: bursts of packets followed by long delays (e.g., three packets quickly, long gap, three more packets)

- High jitter causes problems hearing on phone calls and stutter during live video

Wireless Interference

- Limited frequencies available for access points and clients
- Only one device communicates at a time on a wireless network
- Multiple wireless networks in the same area can create interference
- Recent wireless standards can automatically find best frequencies and avoid those in use
- Can also be manually configured
- Access points can detect interference and move to a better frequency

Frequency Bands

- **2.4 GHz** — limited non-overlapping channels (3 in North America); may not be enough in busy areas
- **5 GHz** — many more frequencies available; better chance of avoiding interference
- **6 GHz** — newest band with many more frequencies to avoid conflicts

Channel Management

- Disable legacy/low-speed support for best throughput — older standards slow down the entire network
- If all devices support the most recent standard, disable legacy options in the access point
- If channels are manually configured, verify they don't conflict with new access points that have appeared
- Consider enabling automatic channel selection so the access point can move to a better channel when available

► EXAMPLE

Channel Overlap (2.4 GHz)

- Non-overlapping channels in the US: 1, 6, 11
- Configuring an access point on channel 8 would overlap with both channel 6 and channel 11
- None of the overlapping devices would have best throughput
- Better to distribute frequencies so they don't interfere (use channels 1, 6, 11)

Managing Interference

- Interference often comes from other access points
- If an access point covers a small area but output power is set to maximum, turn down the power
- Limiting output power reduces interference for other access points in the area

- Split the wireless network into smaller cells with access points separated from each other using different frequencies

Attenuation

- Signal weakens the farther you move from the access point
- Administrators measure attenuation with a Wi-Fi analyzer by walking around the facility
- Solutions to improve signal:
 - Increase power output on the access point (not all support this)
 - Add an external antenna or antenna with additional gain — increases transmission distance with same power
 - Limit cable length between access point and antenna (higher frequencies lose more signal in coax)
 - Check cable for damage that could leak signal

Site Surveys and Heat Maps

- Identify where users access the wireless network and measure signal at those locations
- Check where neighboring organizations' access points are located to avoid interference
- Check what frequencies neighbors use — may open an entire band (e.g., 6 GHz) for you
- Perform multiple site surveys throughout the year — wireless infrastructure changes over time
- **Heat map** — visual representation from a Wi-Fi analyzer showing signal strength
 - Brighter/warmer colors = better signal
 - Cooler colors = worse signal
 - Shows where to add or change access point placement

Client Disassociation Attack

- Denial of service exploiting older unprotected 802.11 management frames
- Attacker can remove devices from the network
- Symptoms: device constantly disconnects and reconnects, or cannot connect at all
- Detection: capture frames using an 802.11 packet capture device to see disassociation frames
- Allows finding and removing/disabling the attacking device
- Prevention: upgrade to latest 802.11 standards which prevent this type of attack

Roaming Misconfigurations

- Multiple access points configured with the same SSID create a large seamless wireless network
- Devices move from access point to access point seamlessly as users roam
- All access points must be configured exactly the same (security features, configuration details)
- If a new access point does not have the same configuration, users are dropped and must reconnect
- Seamless roaming requires identical configuration across all access points

Command Similarity Across Manufacturers

- Commands at the command line are very similar between different manufacturers
- Learning one manufacturer's switch helps manage another manufacturer's switch
- Syntax is not exactly the same, but very close
- Output contains similar information across manufacturers
- Once you learn the technology behind the commands, moving between manufacturers becomes easy

show mac-address-table

- Lists all MAC addresses the switch has learned and which ports they were learned on
- Switch references this table to decide where to send traffic
- Useful for troubleshooting:
 - Why traffic may be sent out every interface
 - Whether the maximum number of entries has been reached

show route

- Lists all routes inside the routing table
- Used to see exactly where routes are for a particular router
- Use on each router along a path to build the full route from source to destination
- Process: find the route that applies to traffic, follow it out the specific interface, repeat on next router

Routing Table Format

- Left side shows abbreviations for route types (e.g., R = RIP, C = connected/directly connected)
- Each route entry shows: destination network, next-hop address, and exit interface
- Evaluate each route to find the most specific match for your traffic

show interface

- Shows status and statistics for a particular interface
- Information includes:
 - Interface status: up, down, connected, or administratively disabled
 - Speed, MTU, and encapsulation types
 - Errors: CRC errors, dropped frames, input/output errors
 - Performance: total number of frames, total broadcasts

- Hardware configuration: duplex, speed (e.g., full duplex, 100 Mbps), media type (e.g., RJ45)
- Input and output rates

show config

- Shows the text-based configuration currently running inside a device
- May also view configurations stored on the storage drive
- Command example: `show running-config` (shows currently active configuration)
- Each manufacturer uses different syntax and format for configuration
- Configuration can be modified at the command line and then viewed again

show arp

- Views the ARP (Address Resolution Protocol) cache inside switches and routers
- Similar to ARP cache in an operating system
- Shows: protocol, IP address, hardware/MAC address, and the interface used to identify that pair
- Useful to determine if a switch or router sees a particular MAC address associated with an IP address

show vlan

- Lists all VLANs on the device and which interfaces are assigned to each VLAN
- Shows:
 - Assigned VLAN IDs
 - Default VLAN for the switch
 - Interfaces that are part of the default VLAN
 - Interfaces specifically assigned to a different VLAN

show power

- Shows PoE power usage for the switch
- Lists all interfaces and whether PoE is on or off for each
- Shows how much power each device is using
- Indicates if additional devices can be added
- Example: switch supports 370W, using 40W, 330W remaining

ping

- Tells you if a device on the network is reachable
- Uses ICMP (Internet Control Message Protocol) to query a device and get a response
- Works the same in macOS, Linux, and Windows
- Output shows: bytes sent/received, sequence numbers, TTL, round-trip time
- If device is unreachable, shows timeout messages
- Use Ctrl+C to stop a continuous ping — provides statistics about the session
- One of the first things to do when troubleshooting — “can you ping that host?”

tracert / traceroute

- Traces the route between your device and the destination
- Maps the entire path, showing every router between source and destination
- Command name: `tracert` (Linux/macOS), `tracert` (Windows)
- Uses ICMP time to live exceeded error messages

How Traceroute Works

1. Sends packet with TTL=1 — first router decreases TTL to 0, sends back TTL exceeded with its IP address
2. Sends packet with TTL=2 — passes first router, second router reports TTL exceeded
3. Continues incrementing TTL until destination is reached
4. Each hop is tested three times by default — output shows three response times per hop

Traceroute Limitations

- Many firewalls filter ICMP messages
- Filtered hops show asterisks instead of statistics
- Windows uses ICMP echo requests; other OSes may allow changing the payload type
- Useful to compare two traceroutes to find where a route is failing

nslookup and dig

- Both query DNS servers and receive responses
- Can look up: canonical names, IP addresses, cache timers, text records, and other DNS records

nslookup

- Included with Windows, Linux, and macOS
- Deprecated — goal is to use `dig` instead
- Shows DNS server used and IP address results for a query

dig

- Included with Linux and macOS
- Windows: available via older `bind` package from isc.org (newer versions no longer include it)
- Shows query details including record type (A record) and responses
- Same information as `nslookup`, different output format

tcpdump

- Captures packets from the command line
- Can view on screen or save to a file
- Included with Linux and macOS; Windows alternative: WinDump
- Can capture all packets or apply filters
- Saves files in `pcap` (packet capture) format — readable by Wireshark and other utilities
- Output shows individual packets with protocol details (IPv4, IPv6, DNS queries, application traffic)

netstat

- Network statistics — shows active network connections to and from your machine
- Common flags:
 - `netstat -a` — show all active connections (incoming and outgoing)
 - `netstat -b` — show Windows executable making each connection
 - `netstat -n` — show IP addresses only (no DNS name resolution)
- Output shows: protocol, local IP address and port, remote IP address and port, connection state (established, closing, waiting)

ipconfig / ifconfig / ip address

- Shows local IP address configuration of a device
- **ipconfig** — Windows
 - Shows IPv6, IPv4 address, subnet mask, default gateway
 - `ipconfig /all` — additional details: device name, node type, MAC address, adapter card, DHCP details, DNS servers
- **ifconfig** — Linux/macOS
 - Specify adapter name (e.g., `ifconfig en0`)
 - Shows MAC address, IPv6 address, IPv4 address, and adapter configuration
- **ip address** — newer Linux command, similar information

arp -a

- Displays the local ARP (Address Resolution Protocol) cache
- Shows IP addresses and their associated MAC addresses on the local subnet
- Works in Linux, macOS, and Windows
- Only shows devices you have recently communicated with
- Pinging a device adds it to the ARP cache
- Useful for looking up a device's MAC address to find it in a switch's MAC address table

Tone Generator and Inductive Probe

- Used to find a single cable in a large bunch of cables
- Tone generator puts an analog tone on the wire
- Inductive probe listens for the tone — only needs to get close to the cable (does not require direct contact)
- Tone generator connects via: modular jack, coax connection, punchdown block, or alligator clips
- Use the probe to move across cables until the tone is identified
- Many tone generator/probe combos also double as a cable tester

Cable Tester

- Checks continuity from one end of a cable to the other
- Verifies pin-to-pin connections (pin 1 to pin 1, pin 2 to pin 2, etc.)
- Identifies:
 - Pins that are not connected
 - Shorts in the cable
 - Crossed wires (e.g., pin 1 connects to pin 3)
- Simple continuity test only — does not show signal capacity or cable category
- Verifies proper punchdown or correct connectors on both ends
- Lights up pins 1 through 8 in order if wired correctly

Network Taps

- Intercepts network traffic and sends a copy to a protocol analyzer or packet capture device
- Physical tap: physically breaks the connection and inserts the tap in the middle
 - May require downtime to install
 - **Passive tap** — not powered; often used for optical fiber
 - **Active tap** — requires additional power to regenerate the signal
- Tap has in/out connections for the network link and separate monitor ports for the analyzer

Port Mirroring / SPAN

- SPAN (Switched Port Analyzer) — a function within the switch itself
- Tells the switch to copy data from any interface and send it to a different interface where a protocol analyzer is connected
- No physical break in the connection needed

- Limitation: limited bandwidth available
 - Example: tapping a link with 2 Gbps total (1 Gbps each direction) but mirror port only has 1 Gbps to the analyzer
- For very large bandwidth uses, a physical tap may be preferred over port mirroring

Wireless Survey Tools

- Gather details about what is going over the air
- Provide information about signal coverage
- Identify potential interference (especially from other access points)
- May be built into the OS or access point, or a third-party tool

Wi-Fi Analyzer

- Provides detailed wireless information; may require specialized hardware
- Sees all 802.11 information going through the air
- Shows: channel strengths, channels in use, access points in the area, potential sources of interference
- Advanced versions include a **spectrum analyzer**:
 - Shows an entire range of frequencies
 - Displays every signal broadcast on those frequencies (802.11 or other sources)
 - Can verify interference from third-party equipment or microwave ovens
- Signal-to-noise ratio visible in analyzer output — narrow ratio indicates weak signal relative to noise; wide separation indicates strong signal

Visual Fault Locator

- A light source specifically designed for optical fiber (like a flashlight for fiber)
- Used to identify breaks in fiber optics
- Light leaks out from around a break, making it visible
- May need to turn off room lights to see clearly
- Can locate faults before installing a fiber patch into the network
- Low-tech but very effective at finding fiber problems

Protocol Analyzers

- Captures frames on a wired or wireless network and presents results in human-readable format
- Capture capability may be built into the device (switch, router, firewall)
- Output can be viewed on the device or exported to a protocol analyzer like Wireshark
- Wireshark provides additional protocol decodes and details about entire conversations
- Some administrators capture all packets to disk for later analysis (big data analysis)
- Useful for finding what part of network communication is performing poorly

Port Scanners (nmap)

- nmap (Network Mapper) — finds open port numbers, operating systems, and application versions
- Active scan — sends queries to a device and examines results
- Capabilities:
 - Discover open ports on a remote device
 - Identify operating system type and version without logging in
 - Identify services running on open ports
 - Scripting engine to extend functionality with custom scripts
- Can scan a single device or an entire IP address range
- Builds a map of IP addresses, operating systems, and services on the network
- Useful for finding rogue devices — can perform Layer 2 scans on the same subnet, making it difficult for rogue devices to hide

► EXAMPLE

nmap scan of 10.1.10.222 — host up with open ports:

- Port 22 — SSH
- Port 80 — HTTP
- Port 443 — HTTPS
- Port 548 — AFP (Apple Filing Protocol)
- Port 2049 — NFS

Unexpected open ports require additional research to identify running services.

Device Discovery (CDP and LLDP)

- Difficult to determine what devices are plugged into a switch, their VLANs, or IP addresses by looking at the front panel
- Discovery protocols gather this information automatically

CDP (Cisco Discovery Protocol)

- Proprietary to Cisco devices
- Provides configuration information without logging in to the device
- Information includes: IP addresses, port identification, software version, native VLAN, hardware-level interface configuration

LLDP (Link Layer Discovery Protocol)

- Vendor-neutral protocol
- Almost all switches support LLDP
- Provides: MAC address, interface identification, switch name, VLAN ID

CDP and LLDP Together

- Both can run simultaneously on a device
- Third-party applications and devices can use these protocols to present information in consolidated form

Speed Test Sites

- Send a lot of information over a link and measure how long it took (download and upload)
- Useful to measure before and after network changes to compare speeds
- Measure at different times of day — more bandwidth available during off hours
- Not all speed test sites are the same:
 - Sites farther away may give different values than closer ones
 - Some sites are constrained by their own bandwidth
- Most accurate results often from your local ISP's speed test site
- Third-party options: Fast.com, SpeedOf.Me, speedtest.net, testmy.net